Progress in Artificial Intelligence (2021) 10:217–243
https://doi.org/10.1007/s13748-021-00238-2

REVIEW

Experimental analysis of machine learning methods for credit score classification

Diwakar Tripathi[1] · Damodar Reddy Edla[2] · Annushree Bablani[3] · Alok Kumar Shukla[4] · B. Ramachandra Reddy[5]

Received: 3 August 2019 / Accepted: 27 February 2021 / Published online: 15 March 2021

Abstract
Credit scoring concerns with emerging empirical model to assist the financial institutions for financial decision-making process. Credit risk analysis plays a vital role for decision-making process; statistical and machine learning approaches are utilized to estimate the risk associated with a credit applicant. Enhancing the performance of credit scoring model, particularly toward non-trustworthy "or non-creditworthy" group, may result incredible effect for financial institution. However, credit scoring data may have excess and unimportant data and features which degrades the performance of model. So, selection of important features (or reduction in irrelevant and redundant features) may play the key role for improving the effectiveness and reducing the complexity of the model. This study presents a experimental results analysis of various combinations of feature selection approaches with various classification approaches and impact of feature selection approaches. For experimental results analysis, nine feature selection and sixteen classification state-of-the-art approaches have been applied on seven benched marked credit scoring datasets.

Keywords Classification · Credit scoring · Feature selection

1 Introduction

Credit scoring is an approach to construct a empirical model to assist the financial institutions for financial decision-making process by considering the risk associated with credit products, and risk with a credit product to a customer is estimated by applying statistical or machine learning (ML) techniques on candidates' historical data [70,104]. According to Thomas et al. [100], "credit scoring is a set of decision models and their underlying techniques that aid credit lenders in the granting of credit" [67,106]. It tries to isolate the impact of various candidates' characteristics based on criminal behavior and defaults. The primary focal point of credit scoring is "to pick whether a credit candidate has a place with financially sound (as trustworthy) or non-trustworthy group." Credit signifies to the amount that is lent by a customer from an institution and it can be estimated by that institution on the basis of customer's credentials like yearly salary, assets and so forth. Various benefits of credit scoring for credit industries include "calculating and reducing credit risk," "making managerial decisions," "cash flow improvement," etc. [76], and its performance is responsible for the profitability of credit industries. It is not a single-stage progression, every so often, financial institutions conduct it in numerous stages [76] and some are as follows:

- *Application Scoring:* It is utilized for the assessment of the authenticity or suspiciousness of new candidates. That evaluation is dependent on social, fiscal and other data assembled at the time of the application.

✉ Diwakar Tripathi
diwakarnitgoa@gmail.com

Damodar Reddy Edla
dr.reddy@nitgoa.ac.in

Annushree Bablani
annushree.bablani@iiits.in

Alok Kumar Shukla
alokjestshukla@gmail.com

B. Ramachandra Reddy
brreddy@iiitdmj.ac.in

1 Thapar Institute of Engineering & Technology Patiala, Patiala, Punjab, India

2 National Institute of Technology Goa, Ponda, India

3 Indian Institute of Information Technology, Sri City, Chittoor, Sri City, Andhra Pradesh, India

4 VIT University AP-Andhra Pradesh, Amaravati, India

5 SRM University AP-Andhra Pradesh, Amaravati, India

– *Behavioral Scoring:* It is comparable as application scoring, though it is for the current customers to inspect and visualize their individual behavior to help dynamic portfolio management processes.
– *Collection Scoring:* It arranges present customers into various categories as per their behavior belongingness, for example, progressively, moderate, no and so on to put attentions on that categories accordingly.
– *Fraud Detection:* In this stage, applicants are classified according to the relative probability that an applicant may be dishonest.

1.1 Motivation

According to the statistics [91] given by Reserve Bank of India (RBI), the users of credit cards during 2012-2016 are depicted in Fig. 1.

1. The number of credit card holders was 24.50 million in the financial year 2015–2016 in India [91].
2. Total number of home loan account in the financial year 2009–2010 was 57.38 lakhs in India [25].

Together with credit cards and home loans, various credit products such as a wide variety of loans, mortgage, mini- & microfinance etc. are also obtainable from financial organizations. Because of huge number of new candidates and existing clients, credit scoring is beyond the realm of imagination to expect to do physically or it requires immense number of specialists with area expertise and practices of client. Right now, credit scoring isn't constrained to banking or credit enterprises just, different areas; for example, telecommunication, real estate and so on are additionally applying credit score forecasting models for investigation of clients' behavior. In this manner, artificial intelligence may defeat the issue of manual credit scoring.

Main focus of credit application scoring is to classify the credit applicant that it belongs to either "legitimate (creditworthy)" and "suspicious (non-creditworthy)" group on the basis of its credentials. So, various researchers have assumed it as a classification problem. Credit scoring models take a vector of attributes for a loan or other credit applicant, and attempt to discriminate between legitimate (creditworthy) or suspicious (non-creditworthy), to discriminate between those that are not likely to default or be in arrears with their payments, and those that are. Enhancing the performance of credit scoring model particularly toward non-trustworthy group may result incredible effect and is also responsible for profitability of financial institution [55,114]. So, selection of important features or reduction in redundant and irrelevant features may improve the effectiveness of credit scoring data classification. This study focuses to present a comparative result analysis and impact of feature section approaches on classification with credit scoring datasets.

Rest of study is organized as follows: Sect. 2 presents a detailed literature survey on credit scoring models; Sect. 3 presents the brief description about credit scoring datasets and performance measure used for comparative results analysis; Sect. 4 presents the previous surveys on credit scoring models; Sect. 5 displays the experimental results of various classification approaches with various feature selection approaches with analysis on seven credit scoring datasets, further followed by conclusion in Sect. 6.

2 Literature survey

This section displays an exhaustive audit on various existing credit scoring models. In most recent two decades, soft computing methods have been broad connected in building powerful and flexible credit scoring models. In this section, significant methodologies designed for credit scoring are presented along with their advantages and limitations; the crucial modeling issues are discussed from the data mining point of view. Figure 2 [67] depicts the frequency of approaches (by considering more than 150 published articles in last decay in peer-reviewed journals and conferences) used in credit scoring models for credit risk evaluation. From Fig. 2, it is observed that hybrid and ensemble approaches are most common methodologies used for credit scoring models.

The majority of the data scientists have considered the credit risk evaluation as classification problem and found it reliable for exploring the hidden patterns in the area of credit scoring data analysis. And, it helps to experts to upgrade their insight for credit risk evaluation. In this specific area, an assortment of ML techniques has been utilized to demonstrate the risk evaluation systems. These strategies are categorized as classification, ensemble classifiers and hybrid classifiers. The classification of these methodologies is appeared in Fig. 3.

From Fig. 3, it is observed that all the approaches for credit scoring (CS) are classified into three categories, namely classification, ensemble classification and hybrid approach, for credit scoring and approaches applied are described in respective Sects. 2.1–2.3. As, we have considered more than 150 published articles in last decay in peer-reviewed journals and conferences and found that credit scoring models have been experimented on four real-world credit scoring datasets, namely Australian Dataset (AUS), Japanese Dataset (JPD), German-categorical Dataset (GCD) and German-numerical Dataset (GND), where AUS & JPD are credit card application datasets and GCD & GND are loan approval datasets, and detailed description about these dataset is presented in Sect. 3.1.

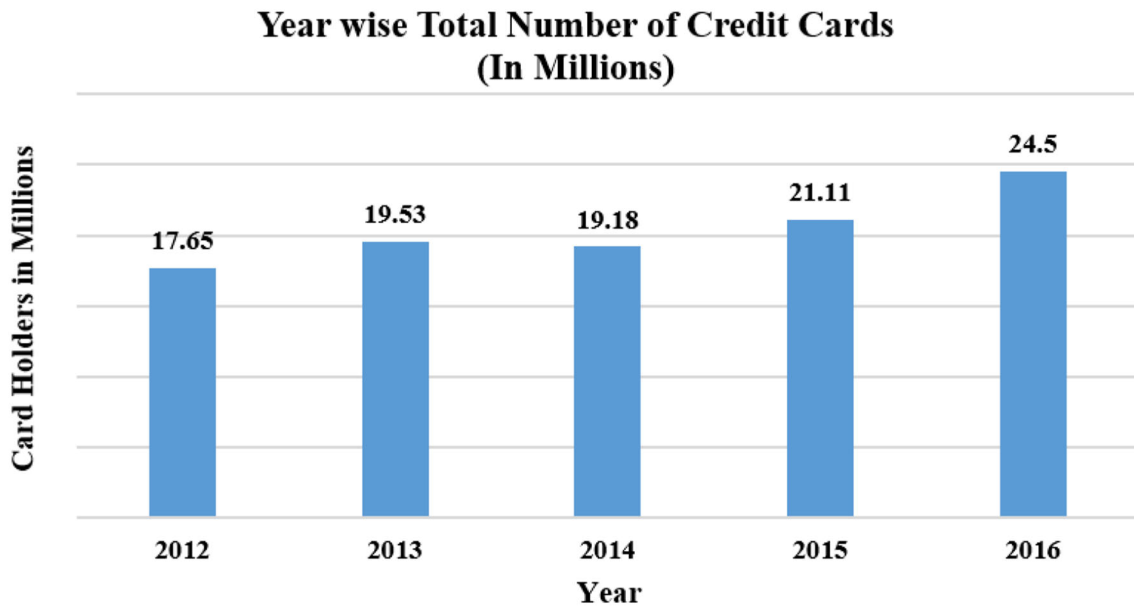

Fig. 1 Rise in issuing of credit cards during 2012–2016

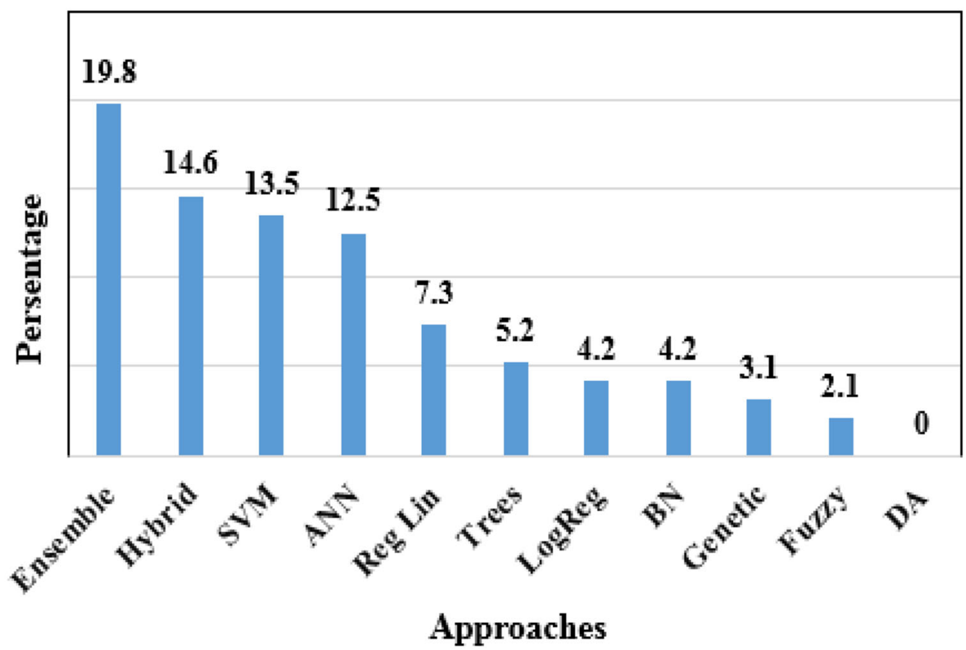

Fig. 2 Comparative graph of various approaches used for credit scoring

2.1 Classification-based CS models

Credit scoring has been considered as binary class classification problem, and various researchers have applied classification approaches for prediction of credit risk associated with an applicant. These approaches are categorized as non-rule- and rule-based approaches, discussed in following subsections.

2.1.1 Non-rule-based CS models

Family of classifiers such as "Artificial Neural Network (ANN)" and "Support Vector Machine (SVM)" have applied. ANN with various learning approaches, SVM as classification tool with various kernel approaches along with some other classifiers have contributed significantly for enhancing the prediction capabilities of credit scoring models. ANNs are influential categorization approach stimulated from human brain capabilities like information processing and learning because of their high ability to learn and iden-

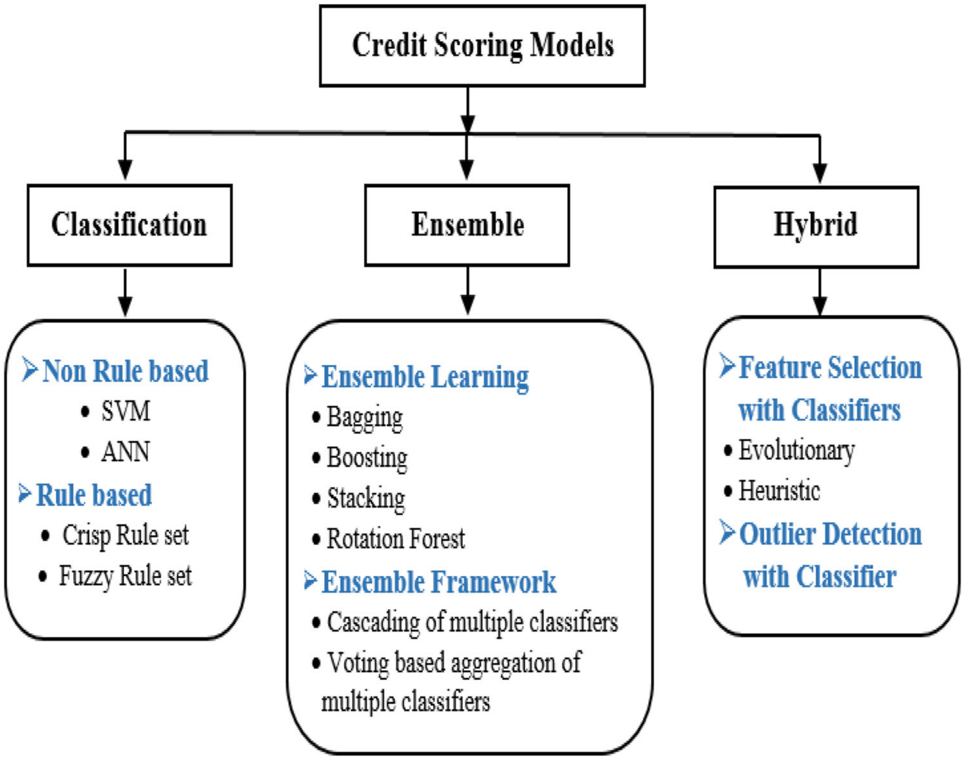

Fig. 3 Taxonomy of various techniques used for credit scoring

tify complex nonlinear associations between input and output [33]. In SVM [17,110] because of "superior features of generalization performance" and "global optimum," many researchers have utilized it as classification approach in various applications. Generally, SVM has better performance as compared to others in most of cases, and as shown in Figure 2, SVM is the third most common approach for credit risk prediction model. Li et al. [61] have proposed a loan evaluation model using SVM with data preprocessing to identify potential applicants for consumer loans. Gestel et al. [109] have utilized "Least Squares Support Vector Machine (LS-SVM)" to analyze the creditworthiness of potential corporate clients. Xiao and Fei [120] have proposed a credit scoring model based on SVM with optimal parameters' values. Yang [122] has proposed an adaptive credit scoring with kernel learning methods. Zhou et al. [129] have applied weighted SVM for credit scoring model, etc.

West [116] has applied various classification approaches such as "Multi-layer Perceptron (MLP)," "Mixture-of-experts," "Radial Basis Function Neural Network (RBFN)," "Learning Vector Quantization (LVQ)" and "Fuzzy Adaptive Resonance" for evaluating the performance of classifiers in terms of accuracy of model. From the experimental results, it is found that Logistic Regression (LR) is the most accurate approach for credit scoring. Atiya [6] has applied external indicators to improve the predictive performance of ANN. Bequé and Lessmann have proposed an approach "Extreme Learning Machines for Credit Scoring: An Empirical Evaluation," and results obtained by proposed approaches are compared with K-nearest Neighbors (KNN), SVM with linear and radial kernels as SVM-L & SVM-R, Classification and Regression Decision Tree (CART), J48, Regularized Logistic Regression (LR-R). And, from the results, SVM has better classification performance [9].

Results obtained from the literature by various classification approaches for credit scoring in terms of classification accuracy with respective dataset and approaches applied along with respective references are depicted in Table 1. As the results are depicted in Table 1, it is observed that SVM-L has the best classification accuracy with AUS and GCD datasets.

2.1.2 Rule-based CS models

One of the major bottlenecks in applying conventional non-rule-based classification approaches, especially as neural network-based classifiers for credit risk prediction model, is that it is hard to translate, in a physically significant manner, in light of the fact that the scholarly learning is numerically encoded in trained synaptic weights. So, various researchers have applied rule-based classifiers for credit risk prediction as follows:

Martens et al. [68] have proposed an approach for rule extraction by SVM; further the results are compared with rule-based classification approach C4.5 and SVM-based classification rule extraction has better classification accu-

Table 1 Non-rule-based classification approaches with performances on credit scoring datasets

Method	Dataset		References
	AUS	GCD	
ANN	84.10	72.80	Bequé and Lessmann [9]
KNN	83.60	66.90	Bequé and Lessmann [9]
SVM-L	87.40	74.80	Bequé and Lessmann [9]
SVM-R	86.10	75.90	Bequé and Lessmann [9]
CART	85.90	55.90	Bequé and Lessmann [9]
J48	84.50	64.10	Bequé and Lessmann [9]
LR-R	86.20	75.40	Bequé and Lessmann [9]
RBFN	87.14	74.60	West [116]
MLP	85.84	73.28	West [116]
LVQ	82.97	68.37	West [116]

racy. Hayashi [31] has utilized "Recursive Rule Extraction approach (Re-Rx)" for assessment of credit risk with both discrete and continuous attributes. A new Ant Colony-based classification rule extraction approach is introduced as $AntMiner^{+}$ [69]. Principle contrasts among $AntMiner^{+}$ and past AntMiner adaptations are the utilization of the better performing ant system, a plainly characterized and enlarged condition for the ants to stroll through, with the incorporation of the class variable to deal with multi-class issues, and the capacity to incorporate interim guidelines in the rule list. Santana et al. [47] have applied Particle Swarm Optimization (PSO) for generating classification rules with combination of neural networks as Self-Organizing Maps (SOM) and LVQ.

Gorzalczany et al. [26] have presented an approach to extract the fuzzy rule using "multi-objective evolutionary optimization algorithms" for financial data. Article [89] presents a "multi-objective genetic (Pittsburgh-type genetic) approach"-based fuzzy rule-based classifiers with chromosome-repairing technique to directly transform the rules, and it permits to find systems with several levels of compromise between its accuracy and interpretability. Hoffmann et al. [36] have proposed two evolutionary fuzzy rule learners: "an evolution strategy that generates approximate fuzzy rules" and "a genetic algorithm that extracts descriptive fuzzy rules with boosting procedure" to identify fuzzy rule that best matches in training samples. Similarly, Yu et al. [125] have proposed a rule extraction approach based on agent-based fuzzy group decision-making with multi-criteria decision analysis.

Results obtained by various approaches based on rule-based classification for credit scoring in terms of classification accuracy with respective dataset and approaches applied along with respective references are depicted in Table 2. From the results as shown in Table 2, it is observed that Fuzzy rule based on Multi-Objective Genetic Algorithm (MOGA) has the best classification accuracy with AUS and JPD datasets and second best with GCD dataset; agent-based fuzzy group decision approach has achieved the best classification accuracy with GCD dataset.

2.2 Ensemble classification-based CS models

Various rule-based such as PART, C4.5 and DT and non-rule-based classification approaches such as SVM and ANNs have been successfully applied to learn various problems up to this point. However, each classifier cannot perform well for all problems. Usually, they perform well for some specific problems. As a consequence, ensemble could be a robust approach to induce concerning best classifier for any problem [77]. This methodology fortifies the classifiers in mistake inclined subspaces, and subsequently, it can prompt higher execution for characterization. Generally, the result of combination of diverse classifiers is better than individual classifier [4,8]. Numerous studies uncovered that individual classifier indicates moderate performance when contrasted with ensemble ones. Various approaches have been developed for constructing the ensembles, and these approaches can be categorized in two types as ensemble learning and ensemble classifiers framework. Ensemble Learning refers to "The procedures employed to train multiple learning machines and combine their outputs, treating them as a committee of decision makers" with principle that the committee decision may be more accurate and robust than any individual member. Ensemble classifiers framework (Multi-Classifier Systems) focuses on "The combination of classifiers forms heterogeneous or homogeneous modeling backgrounds to give the final decision."

Ensemble learning approaches are Bagging [71], Boosting [52], Stacking [50], Rotation Forest [83], etc. Basically, these approaches focus on modifying the training samples with motivation to improve predictability of model by considering various sets from the training set, elimination of redundant samples in training set, manipulation of input features in training set, injecting randomness in training set, etc. Further, these sets are fed to same classifiers, and outputs of classifiers with different sets are aggregated by aggregation approach (mostly by majority voting). Zhang et al. [127] have applied "Vertical Bagging Decision Tree Model (VBDTM)" for credit scoring. Nanni and Lumini [73] have presented a comparative analysis on ensemble classifiers for bankruptcy prediction; from the experimental observation, it is concluded that random subspace with "Levenberg–Marquardt Neural Nets (LMNC)" has the best performance as compared to rest of the approaches. Xia et al. [119] have proposed an approach as Bstacking by combining Bagging with the Stacking.

Ensemble framework for classification considers various classifiers on same set of training dataset (without altering training samples). These classifiers can be aggregated

Table 2 Rule-based classification approaches with performances on credit scoring datasets

Method	Dataset AUS	JPD	GCD	GND	References
SOM + PSO	85.54	–	69.84	–	Santana et al. [47]
SOM + varPSO	85.36	–	70.20	–	Santana et al. [47]
LVQ + PSO	86.41	–	68.23	–	Santana et al. [47]
LVQ + varPSO	85.43	–	69.81	–	Santana et al. [47]
PART	73.58	–	69.90	–	Santana et al. [47]
Fuzzy rule based on MOGA	89.1	89.0	78.5	–	Gorzałczany et al. [26]
Continuous Re-RX algorithm	86.93	–	75.22	–	Hayashi [31]
Re-RX with J48graft	86.06	–	72.78	–	Hayashi [31]
Sampling Re-RX	86.48	–	73.2	–	Hayashi [31]
Re-RX	86.06	–	71.82	–	Hayashi [31]
SPEA2 without SFP	–	85.51	70.70	–	Rudzinski [89]
SPEA2 with SFP	–	86.38	72.40	–	Rudzinski [89]
NSGA-II without SFP	–	85.22	71.40	–	Rudzinski [89]
NSGA-II with SFP	–	84.06	72.90	–	Rudzinski [89]
Agent-based fuzzy group decision	–	86.16	82.00	–	Yu et al. [125]
Neurofuzzy + Boosting	86.7	–	76.30	–	Hoffmann et al. [36]
$AntMiner^{+}$	84.05	–	71.88	–	Martens et al. [69]
AntMiner	84.09	–	71.08	–	Martens et al. [69]
AntMiner-2	84.30	–	70.66	–	Martens et al. [69]
AntMiner-3	83.61	–	71.29	–	Martens et al. [69]
Ripper	84.52	–	73.18	–	Martens et al. [69]
SVM	85.7	–	–	–	Martens et al. [68]
C4.5	84.2	–	–	–	Martens et al. [68]

by various ways such as single & multilayer ensemble and cascading of multiple classifier framework [51,80]. In case of single layer ensemble classifier framework, various base classifiers outputs are consolidated by aggregation approach, whereas, with multi-layer, base classifiers are segregated in hierarchical manner and in each layer's classifiers are consolidated, further fed to next layer for consolidation. In each layer, outputs by base classifiers are consolidated by aggregation approach, where in cascading of multiple classifier framework, output obtained by base classifiers is fed to another classifier for consolidation. From the literature, it is clear that there are two types of ensemble frameworks, i.e., "homogeneous" and "heterogeneous" ensemble classifiers frameworks [7,8]. In "homogeneous ensemble classifiers framework," the base classifiers of same kind are utilized, whereas "heterogeneous ensemble classifiers framework" made out of base classifiers that have a place with different types. Most and well-known approaches to consolidate the base classifiers are Unanimous Voting (UV) [62], Majority Voting (MV) and Weighted Voting (WV) [3,4].

A multi-layer ensemble framework with heterogeneous classifiers overcomes the limitations of conventional performance bottlenecks [7,8]. Edla et al. [21] have presented a comparative analysis of various ensemble frameworks, in which various classifiers are ensembled with MV and WV in layered and non-layered manner. From results, it is observed that multi-layered approach with WV as aggregation approach (LWV) has better performances as compared to rest of the combinations, and it improves the performances as compared to its base classifiers [21,102,103]. And, from the experimental outcomes, it is observed that optimized classifiers placement such as best classifiers at last layer has significant improvement toward classification performance as compared to random placement of classifiers in layered manner [103]. Guo et al. [28] have proposed "Multi-Stage Self-Adaptive Classifier Ensemble (MSEC)"; in this approach optimized base classifiers are considered for modeling classifier ensemble.

Various ensemble approaches applied for credit risk prediction in terms of predictive accuracy with respective dataset along with respective references are depicted in Table 3. From the results as shown in Table 3, it is observed that Neighborhood Rough Set (NRS)-based feature selection with LWV (with three heterogeneous classifiers as MLFN, NB and QDA at first layer and in second layer two homogeneous ANN-based classifiers) approach has the best classification accuracy with AUS and GCD datasets.

Table 3 Ensemble classification approaches with performances on credit scoring datasets

Method	Dataset				References
	AUS	JPD	GCD	GND	
EFS + LWV	92.69	89.06	85.18	84.69	Tripathi et al. [103]
MSEC	87.40	87.00	78.30	–	Guo et al. [28]
NRS + LWV	95.39	–	86.47	–	Tripathi et al. [102]
Bstacking	88.28	–	78.66	–	Xia et al. [119]
FS + WV	87.32	87.98	77.12	–	Tripathi et al. [105]
LWV	92.58	89.16	85.82	85.62	Edla et al. [21]
Consensus hybrid ensemble	88.10	88.70	79.00	–	Ala'raj and Abbod [4]
Consensus system approach	88.98	87.88	77.72	–	Ala'raj and Abbod [3]
RF + CTD	86.10	86.40	75.20	–	Abellán and Castellano [2]
RS-bagging DT	88.17	–	78.52	–	Wang et al. [114]
Boosting (LRA)	86.56	–	76.14	–	Wang et al. [113]
VBDTM	91.97	–	81.64	–	Zhang et al. [127]
Random subspace + LMNC	87.05	87.34	73.93	–	Nanni and Lumini [73]
ANN classifiers	87.25	85.91	76.60	–	Tsai and Wu [108]

2.3 Hybrid approach for CS

A hybrid approach is a dynamic organization with association of the concepts for problem solving steps of numerous methods in the direction of the enhancement of a specific consequence. It is a strong approach to improve the performance of credit scoring model. So, it also got second most attention by various researchers in the area of credit scoring (as shown in Fig. 2). As in credit scoring datasets have large number heterogeneous features such as categorical, numerical with small range (such as age in between 20 and 80) and numerical with huge range (such as annual income) and there may be some redundant or irreverent features which can reduce the model's predictive performance. So, in order to improve the model's predictive performance, various scholars have applied various data preprocessing step such as feature selection and outlier detection.

Feature selection is combinatorial optimization problem and selects best set of features from actual set of features, and can be categorized as wrapper, filter and embedded methods. In wrapper method, the feature selection process is based on a specific classifier and evaluates all the possible combinations of features against the evaluation criterion [38,95]. But filter methods do not consider a specific classifier for feature selection. Embedded method is also a classifier-dependent approach and embeds the features during model building process. Bio-inspired approaches toward problem-solving seems almost ideal because of properties such as "self optimization," "flexibility" and "simple set of ground rules" [14,74,96] and it got more attention by various researcher for features selection. Chi et al. [16] have presented a dual credit scoring model by combining GA-based feature selection approach for important features to combine the bank's internal behavioral with the external credit bureau for credit scoring. Oreski and Oreski [75] have applied GA-based feature selection with ANN to enhance the classification performance of credit scoring model. Rough set and Tabu search with conditional entropy as criteria function to investigate for most favorable set of features have proposed by Wang et al. [115]. Binary Particle Swarm Optimization (BPSO) to investigate for most favorable set of features with SVM for credit score classification has been applied by Huang and Dun [40]. Hybridization of Gravitational Search Algorithm (GSA) with PSO in order to improve the exploration and exploitation capabilities has been utilized for selecting most favorable set of features [21]. Zhang et al. [128] have proposed "Multi-stage Hybrid model with enhanced multi-population Niche Genetic Algorithm (MHNGA)" with consideration of optimal feature subset and optimal classifier subset to improve the prediction performance of credit scoring. A hybrid model of " Generalized Fuzzy Soft Sets (GFSS) theory-based ensemble classification" with "adaptive elastic net-based feature selection" is presented by Xu et al. [121].

Liang et al. [63] have applied various filter and wrapper-based feature selection methods on various diverse prediction models for the credit scoring data classification. Clustering of features on the basis of correlation coefficient has been presented [105]. Ping and Yongheng [78] have applied "Neighborhood Rough Set (NRS)"-based feature selection with SVM for credit scoring. Hu et al. [37] have presented a NRS model to resolve the problem of heterogeneous feature subset selection. Some more hybrid models as two-stage modeling procedure (with first stage as feature selection and second as classification) are also presented for credit scoring such as "Multivariate Adaptive Regression Splines (MARS)" and "F-score"-based feature selection and SVM-based clas-

Table 4 Hybrid approaches with performances on credit scoring datasets

Method	Dataset				References
	AUS	JPD	GCD	GND	
MHNGA	87.54	87.20	76.82	–	Zhang et al. [128]
GFSS	88.10	89.40	87.60	–	Xu et al. [121]
SR + ANN	84.09	–	—	–	Wongchinsri and Kuratach [118]
GA + NB	85.56	–	–	74.03	Liang et al. [63]
LDA + MLP	86.00	–	–	73.44	Liang et al. [63]
HGA-NN	–	–	78.90	–	Oreski and Oreski [75]
RS + TS + LR	–	86.40	–	–	Wang et al. [115]
Sampling + F-score + SVM	86.76	–	76.84	–	Hens and Tiwari [35]
IFS	90.90	–	–	80.20	Liu et al. [66]
NRS + SVM + Grid search	87.52	–	76.60	–	Ping and Yongheng [78]
NRS-based FS	–	85.48	74.50	–	Hu et al. [37]
SVM + Grid search	85.51	–	76.00	–	Huang et al. [39]
SVM + Grid search + F-score	84.20	–	77.50	–	Huang et al. [39]
SVM + GA	86.90	–	77.92	–	Huang et al. [39]
GA + SVM	90.19	–	84.24	–	Huang and Wang [41]

sification in articles [15,35], respectively; "Stepwise Regression (SR)," "MARS" and "discriminant analysis" approaches for selecting the significant predictor variables with ANN as classification approach are combined for credit scoring in articles [59,60,118], respectively. Huang et al. [39] have presented an hybrid approach with various strategies, namely Grid, Grid with F-score and GA with SVM.

Various hybrid approaches applied for credit risk prediction in terms of accuracy with respective dataset and references are depicted in Table 4. From the results as shown in Table 4, it is observed that GA-based feature selection and SVM-based classification have the best classification accuracy with AUS and GCD datasets.

As shown in Fig. 2, it is displayed that hybrid approach has the second most highest frequency and most of the researchers have found the feature selection as prominent way to improve the classification performances. By surveying the articles published in last decay, the approaches have been applied for credit scoring as displayed in Tables 1, 2, 3 and 4. Apart from these approaches, some other feature selection and classification approaches are not yet applied for credit scoring as follows. Infinite Latent Feature Selection (ILFS) [86] is probabilistic latent graph-based feature ranking approach by taking into consideration "all the probable subsets of features, as paths on a graph, bypassing the combinatorial problem analytically." Laplacian Score-based Feature Selection (LS) [34] is basically dependent on Laplacian Eigen-maps and Locality-Preserving Projection. Relief [53,54] is based on the "recognition of feature value differences between nearest neighbor instance pairs." Unsupervised Discriminative Feature Selection (UDFS) [123] is a methodology which chooses the most discriminative feature subset from genuine list of features. In Feature Selection via Concave Minimization (FSV) [10], it considers a plane is built to such an extent a weighted total of separations of misclassified points. Multi-Cluster-based Feature Selection (MCFS) [13] is an unsupervised feature selection approach, and it selects the features with the multi-cluster structure of the data which can be best preserved. Correlation Coefficients-based Feature Ranking (CFS) evaluates subsets of features on the basis of the hypothesis: "Good feature subsets contain features highly correlated with the classification, yet uncorrelated with each other" [29,93]. Feature Selection via Eigenvector Centrality (ECFS) [84,85] is a graph-based feature selection approach that ranks features on the basis of graph centrality measure (Eigenvector centrality). Feature selection by Local Learning-based Clustering (LLCFS) [126] allocates weight to each feature and integrates it into the inherent regularization of the LLCFS algorithm to consider the importance of each element for the bunching.

Some classification approaches are summarized below. Logistic Regression (LR) [27,30,58] can be considered as an extraordinary case of linear regression models by violation of normality assumptions. Multilayer Perceptron (MLP) is a feed-forward neural network with at least three layers input, hidden and output layer. Other than input layer, neurons at each layer use a nonlinear activation function with back-propagation for training [32,88,90]. Radial Basis Function Neural Network (RBFN), similar to MLP, RBFN [12,49], is a feed-forward architecture with single hidden layer and radial basis function as an activation function. Extreme Learning Machine (ELM), similar to other ANNs such as MLP, ELM [44,46] as a classification approach structured in three layers, with difference assigns random weights

Table 5 Description of credit scoring datasets

S. No	Dataset	Total # Instances	# Instances in Class1/Class2	# Features	Cate/Num # Features	# Classes
1	Australian (AUS)	690	307/383	14	8/6	2
2	Bankruptcy	250	143/107	6	0/6	2
3	Bank-marketing	4521	4000/521	16	9/7	2
4	Japanese (JPD)	690	307/383	15	9/6	2
5	German-categorical (GCD)	1000	700/300	20	13/7	2
6	German-numerical (GND)	1000	700/300	24	0/24	2
7	Taiwan	30000	23365/6635	23	0/23	2

#, number of; Cate, categorical, Num, numerical

Table 6 Confusion matrix parameters definition

Abbreviation	Explanation
TP	Actual creditworthy cases which are classified as creditworthy
TN	Actual non-creditworthy cases which are classified as non-creditworthy
FP	Actual creditworthy cases which are classified as non-creditworthy
FN	Actual non-creditworthy cases which are classified as creditworthy

and biases in between input an hidden layers and weights in between hidden and output layers are estimated using weights (between input and hidden layers), biases and data samples [42,43,45]. Huang et al. [45] have proposed equality constrained optimization-based faster ELM, which are as follows:
Faster-1 ELM (ELM-1):- If the number of training samples is not huge.

$$b = H^T \left(\frac{I}{C} + HH^T \right)^{-1} T \quad (1)$$

Faster-2 ELM (ELM-2):- If the number of training samples is huge.

$$b = \left(\frac{I}{C} + H^T H \right)^{-1} H^T T \quad (2)$$

where I represents an identity matrix of size number of hidden neurons and C is regularization factor.

Time Delay Neural Network (TDNN) [112] is a multilayer neural network which classifies patterns with shift-invariance "that does not require explicit segmentation prior to classification" and model context "higher layer receives input not only from the layer along with activations/features" at each layer of the network [57,94]. Linear Discriminant Analysis (LDA) [20,22] as a classification approach is utilized to find a linear combination of features that characterizes into the classes of objects. Naive Bayes (NB) [48] is a probabilistic learning model based on the Bayes theorem for classification and can be applied on data with both continuous and categorical features. Decision Tree (DT) [87] constructs a tree for learning as a predictive model with Nodes as "Test for the value of a certain attribute"; Edges or Branches as " Correspond to the outcome of a test and connect to the next node or leaf" and Leaf nodes as "Terminal nodes that predict the outcome." Random forest (RF) [11] is an approach for classification and regression problem in association with ensemble learning and operated by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes. And, it is a better way for decision trees in case of over-fitting problem [24]. K-nearest Neighbor (KNN) [5] is a classification approach of a type of lazy learning, in which the function is only approximated locally and all computation is deferred until classification and it is based on learning by analogy. Partial Decision Tree (PART) [23,117] combines two approaches "C4.5" and "Ripper" to avoid their respective problems with avoidance of global optimization for constructing a rule set and work in "separate-and-conquer manner." Support Vector Machine for separation of a set of feature vector with different classes' membership by the supervised machine learning, SVM, constructs an optimal decision plane [92,111]. Sequential Minimal Optimization (SMO) has better generalized classification performance with "slower," "complex" and "subtle training" [79], because it requires "expensive third-party quadratic programming (QP) solvers" [82] for training.

Table 7 Results on Australian dataset

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	LLCFS	CFS
LDA	Acc	85.96	86.68	85.13	85.96	85.15	84.83	85.96	85.39	84.83	86.81
	Sen	84.23	84.19	84.37	86.73	88.12	82.44	86.97	86.23	86.54	85.88
	Spe	87.97	88.23	88.54	83.88	84.23	88.19	85.37	85.73	84.43	89.16
	G-M	86.08	86.19	86.43	85.29	86.15	85.27	86.17	85.98	85.48	87.50
NB	Acc	79.89	79.86	79.15	80.15	79.59	78.34	86.69	86.25	78.34	80.88
	Sen	81.78	81.88	82.11	81.03	81.85	81.00	85.30	85.12	81.20	83.13
	Spe	80.05	79.01	78.00	80.22	78.98	76.95	88.29	87.85	77.14	78.97
	G-M	80.91	80.43	80.03	80.62	80.40	78.95	86.78	86.48	79.15	81.02
TDNN	Acc	90.37	92.05	90.14	91.13	90.25	89.88	91.89	92.07	89.79	91.80
	Sen	90.77	89.98	91.14	91.64	90.85	89.91	90.51	92.87	90.14	92.27
	Spe	91.68	94.48	92.05	92.56	91.76	90.81	91.41	93.79	91.04	93.20
	G-M	91.22	92.20	91.59	92.10	91.30	90.36	90.96	93.33	90.59	92.73
KNN	Acc	72.34	75.21	71.70	73.06	70.89	71.48	81.20	73.74	71.48	73.83
	Sen	73.06	72.96	72.42	71.79	71.60	72.19	82.01	74.48	73.19	74.57
	Spe	72.33	75.20	71.69	73.05	70.88	71.47	81.19	73.73	71.47	73.82
	G-M	72.70	74.58	72.05	72.42	71.24	71.83	81.60	74.10	72.32	74.19
DT	Acc	83.29	85.26	82.34	82.47	82.87	81.33	85.37	86.54	81.18	83.03
	Sen	83.18	86.48	82.45	84.01	84.71	82.19	85.37	88.11	82.19	84.90
	Spe	85.42	88.07	83.95	81.09	81.59	82.27	86.96	86.34	82.27	81.94
	G-M	84.30	87.27	83.20	82.54	83.13	82.23	86.17	87.22	82.23	83.41
ELM	Acc	81.21	81.45	80.80	82.48	81.06	82.58	83.17	83.44	82.17	82.76
	Sen	82.14	83.34	82.07	82.03	82.15	82.40	86.55	85.51	83.17	83.87
	Spe	82.41	80.19	80.18	83.71	80.97	81.60	82.42	82.27	81.60	80.19
	G-M	82.28	81.75	81.12	82.86	81.56	82.00	84.46	83.88	82.38	82.01
ELM-1	Acc	81.17	82.35	81.10	81.06	81.18	81.42	85.52	84.50	82.18	82.88
	Sen	83.10	82.57	83.93	83.69	82.37	83.82	85.30	84.72	83.21	82.83
	Spe	82.27	81.74	81.11	82.86	81.55	81.99	84.45	83.87	82.37	82.00
	G-M	82.68	82.15	82.51	83.27	81.96	82.90	84.87	84.29	82.79	82.42
ELM-2	Acc	82.45	84.29	82.11	82.28	80.82	83.52	85.22	84.80	82.20	83.32
	Sen	82.68	84.67	82.32	80.66	81.87	82.40	84.86	84.28	82.78	82.41
	Spe	83.51	85.52	83.15	84.11	82.78	83.23	85.72	85.13	83.62	83.24
	G-M	83.09	85.09	82.74	82.36	82.32	82.81	85.29	84.71	83.20	82.82

3 Credit scoring datasets and performance measure

3.1 Datasets

Because of the privacy issues, financial institutions or banking systems are not disclosing the real time dataset. To make a comprehensive comparison on credit scoring, we have reviewed last 10 year published articles in this area and used the public datasets mentioned on these articles. From the literature, seven datasets such as Australian, Bank-marketing, Bankruptcy, Japanese, German-categorical, German-numerical and Taiwan datasets are publicly available. And, out of seven datasets, Australian and German-categorical datasets are the most popular datasets and more than 75% articles are experimented on these two datasets. Detailed description of aforementioned credit scoring datasets obtained from UCI data repository [64] used in this research work is tabulated in Table 5. All the datasets are real-world credit scoring datasets with different application such as loan, and credit card and due to privacy issues some feature values on some dataset are altered by symbolic representation.

Australian [98] and Japanese [19] datasets are related to credit card applications and both have categorical and numerical attributes along with some missing values. In Australian dataset, the categorical attributes "1, 4, 5, 6, 8, 9, 11 and 12" have "2, 3, 14, 9, 2, 2, 2 and 3" values, respectively, and rest

Table 7 continued

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	LLCFS	CFS
RF	Acc	87.10	88.36	87.94	87.66	86.30	86.98	85.12	86.21	84.98	87.34
	Sen	87.97	89.24	88.82	88.54	87.16	87.85	85.97	87.07	85.83	88.21
	Spe	88.06	89.33	88.91	88.63	87.25	87.94	86.06	87.16	85.92	88.30
	G-M	88.01	89.29	88.86	88.58	87.21	87.89	86.01	87.12	85.87	88.26
PART	Acc	83.62	84.05	83.76	84.05	83.26	83.52	84.05	85.51	82.52	83.47
	Sen	83.45	83.88	83.59	83.88	83.09	81.35	83.88	82.34	82.35	83.30
	Spe	84.20	84.30	84.01	84.30	83.51	84.77	84.30	86.67	83.01	84.05
	G-M	83.83	84.09	83.80	84.09	83.30	83.04	84.09	84.47	82.68	83.68
MLP	Acc	84.92	85.50	84.49	85.51	84.12	83.40	84.78	85.65	83.40	85.36
	Sen	85.01	85.39	83.53	86.21	83.64	83.52	88.21	87.63	83.64	85.45
	Spe	86.26	87.74	86.39	87.23	85.98	84.69	85.01	85.94	84.19	85.88
	G-M	85.63	86.56	84.95	86.72	84.80	84.10	86.59	86.78	83.92	85.66
RBFN	Acc	82.75	83.62	81.33	84.05	81.30	81.94	85.94	85.79	81.94	83.62
	Sen	84.43	86.01	83.78	86.07	82.92	83.63	88.24	88.35	83.74	85.74
	Spe	82.24	83.12	80.94	83.88	81.23	81.20	85.91	85.86	81.15	82.87
	G-M	83.33	84.55	82.35	84.97	82.07	82.40	87.07	87.10	82.44	84.29
SVM-P	Acc	84.71	85.52	84.92	85.93	85.60	85.60	87.07	85.59	85.60	86.13
	Sen	84.27	85.63	84.95	85.54	85.61	84.98	86.66	85.17	85.37	86.09
	Spe	87.74	88.60	87.59	88.41	88.19	88.48	90.20	88.70	87.85	88.71
	G-M	85.99	87.10	86.26	86.96	86.89	86.71	88.41	86.92	86.60	87.39
SVM-R	Acc	85.50	86.17	85.87	86.81	87.42	87.98	88.36	88.63	85.07	87.56
	Sen	84.21	84.51	84.72	85.25	86.58	86.40	86.76	86.87	83.64	85.93
	Spe	88.36	88.20	88.23	89.30	89.69	90.30	90.99	91.31	87.74	90.37
	G-M	86.26	86.34	86.46	87.25	88.12	88.33	88.85	89.06	85.66	88.12
LRA	Acc	86.95	86.95	86.50	86.37	85.59	85.94	86.52	86.08	85.98	87.79
	Sen	84.86	84.86	84.15	83.58	83.69	83.96	84.77	83.97	84.09	85.46
	Spe	87.56	88.50	87.16	86.19	85.82	86.38	87.36	86.90	86.28	87.67
	G-M	86.20	86.66	85.64	84.87	84.75	85.16	86.05	85.43	85.18	86.56
SMO	Acc	85.50	85.50	84.20	85.50	84.73	84.08	86.29	86.53	84.18	86.32
	Sen	85.35	85.49	84.19	83.84	83.26	82.03	85.22	84.97	84.39	84.79
	Spe	86.37	86.37	85.06	86.36	85.61	85.94	87.18	87.42	85.04	87.22
	G-M	85.86	85.93	84.62	85.09	84.43	83.96	86.20	86.19	84.71	86.00

are numerical attributes. Similarly, in Japanese dataset the attributes "1, 4, 5, 6, 7, 9, 10, 12 and 13" have "2, 4, 3, 14, 9, 2, 2, 2 and 3" categorical values, respectively, and rest are numerical attributes.

Bankruptcy dataset [81] contains 250 samples with 6 features along with 1 class label. These six features represent "industrial risk," "management risk," "financial flexibility," "credibility," "competitiveness" and "operating risk" with feature values as positive, average or negative. In this dataset, class label represents that sample is bankruptcy or non-bankruptcy with 143 and 103 samples of bankruptcy and non-bankruptcy class, respectively.

Bank Direct Marketing (Bank-marketing) dataset [72] is connected with "direct marketing campaigns of a Portuguese financial institution," and it was based on phone calls. Frequently, more than one contact to a similar customer was required, so as to get to if the item (bank term deposit) would be (or not) bought in. This dataset has 45211 with 16 features along with a class label (two classed); to make it as small dataset it is also provided to test more computationally demanding machine learning algorithms (e.g., SVM) with 4521 samples with 16 feature as bank dataset. Bank dataset is generated by the same author by considering 10% samples of each class. In this dataset, age, balance, days, duration, campaign, pdays and previous are the numerical features. Where, days represents "last contact day of the month" , campaign as "number of contacts performed during this campaign and for this client," pdays as "number of days that passed by

Table 8 Results on Bankruptcy dataset

	Measures	ALL	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	LLCFS	CFS
LDA	Acc	99.16	99.58	98.95	99.33	98.83	99.58	98.95	99.86	98.12	98.83
	Sen	99.82	99.68	98.96	99.12	99.35	99.58	99.41	99.86	98.39	99.47
	Spe	99.68	99.72	99.08	99.34	99.29	99.58	98.99	99.86	98.73	99.45
	G-M	99.75	99.71	99.02	99.23	99.32	99.58	99.20	99.86	98.56	99.46
NB	Acc	97.62	97.20	96.12	98.03	97.12	97.20	98.45	98.03	98.45	97.12
	Sen	98.32	98.23	96.63	98.38	98.90	97.55	98.76	98.52	98.97	97.67
	Spe	98.40	98.06	98.03	98.97	97.85	98.18	98.64	98.87	99.10	97.76
	G-M	98.36	98.15	97.33	98.68	98.38	97.87	98.70	98.69	99.03	97.71
TDNN	Acc	98.76	99.68	98.73	98.54	98.15	98.46	98.57	98.21	98.57	98.90
	Sen	99.14	99.68	99.59	99.10	98.54	99.39	99.53	98.34	99.31	99.26
	Spe	98.86	99.69	99.28	99.53	98.30	99.37	99.54	98.62	99.26	99.37
	G-M	99.00	99.68	99.44	99.31	98.42	99.38	99.54	98.48	99.28	99.32
KNN	Acc	99.58	98.50	98.77	98.75	97.66	97.53	97.08	98.75	98.33	98.66
	Sen	99.59	99.18	99.26	99.50	98.20	98.39	97.88	99.53	98.61	99.04
	Spe	99.86	99.04	99.22	99.31	99.13	98.43	98.23	99.07	98.57	98.68
	G-M	99.72	99.11	99.24	99.40	98.66	98.41	98.05	99.30	98.59	98.86
DT	Acc	97.08	96.66	96.95	97.78	96.12	97.66	98.03	96.78	98.45	97.12
	Sen	99.55	96.79	97.29	98.57	96.97	98.15	98.56	97.58	99.20	97.68
	Spe	97.89	97.66	97.58	98.44	96.41	98.15	98.95	97.56	98.62	97.77
	G-M	98.71	97.22	97.44	98.50	96.69	98.15	98.75	97.57	98.91	97.73
ELM	Acc	99.88	99.88	99.12	99.58	98.08	99.88	99.88	99.88	99.88	99.08
	Sen	99.91	99.82	99.72	99.61	98.14	99.47	99.82	99.02	99.87	99.80
	Spe	98.78	98.88	99.28	99.59	98.88	99.86	99.18	99.85	99.12	99.64
	G-M	99.34	99.35	99.50	99.60	98.51	99.66	99.49	99.43	99.34	99.72
ELM-1	Acc	99.91	99.95	99.29	99.82	99.01	99.91	99.87	99.91	99.93	99.23
	Sen	99.12	99.95	99.83	99.46	99.69	99.32	99.45	99.54	99.91	99.97
	Spe	99.92	99.34	99.68	99.17	99.83	99.88	99.74	99.82	99.19	99.49
	G-M	99.56	99.67	99.91	99.32	99.76	99.60	99.59	99.68	99.55	99.72
ELM-2	Acc	98.75	97.91	90.53	97.91	95.83	99.16	98.45	99.58	98.45	95.83
	Sen	99.41	98.23	95.08	98.90	96.53	99.17	98.57	99.65	98.62	96.82
	Spe	99.32	98.65	91.23	98.03	96.47	99.78	98.96	99.36	99.02	96.12
	G-M	99.36	98.44	93.14	98.46	96.50	99.47	98.77	99.50	98.82	96.47

after the client was last contacted from a previous campaign" and previous as "number of contacts performed before this campaign and for this client," respectively. And, rest of the features such as job, marital status, education, default, housing, loan, contact, month and poutcome as "outcome of the previous marketing campaign" are categorical attributes with "12, 7, 4, 2, 2, 2, 3, 12, 4" values, respectively.

Taiwan dataset is of an important bank (a cash and credit card issuer) in Taiwan and the targets are credit card holders of the bank [124]. In this dataset, all features are numerical features. Amount "it includes both the individual consumer credit and his/her family (supplementary) credit," Gender "male as 1 and female as 2," Education "graduate, school, university, high school, others as 1, 2, 3, 4, 5, respectively," Marital status "married, single, others as 1, 2, 3, respectively," Age (in years). Features 6 - 11: History of past payment "tracked the past monthly payment records (in previous 6 months as 1 for respective month payment delayed by 1 month)." Features 12–17: Amount of bill in previous 6 months. Features 18–23 present the amount of previous payment paid in previous 6 months.

The German-categorical dataset [99] has 1000 samples with 21 attributes, each samples contains 13 nominal, 7 numeric attributes and 1 class attribute which indicates that is approved or rejected, where 700 samples are approved applicants and 300 samples are rejected. Attributes "1, 3, 4, 6, 7, 9, 10, 12, 14, 15, 17, 19 and 20" are categorical with "4, 5, 11, 5, 5, 5, 3, 4, 3, 3, 4, 2 and 2" values, respectively, while the rest

Table 8 continued

	Measures	ALL	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	LLCFS	CFS
RF	Acc	99.63	99.26	98.89	99.88	98.86	99.69	99.88	99.88	99.88	99.96
	Sen	99.19	99.54	99.12	99.53	99.58	99.11	99.74	99.34	99.82	99.82
	Spe	99.12	99.86	99.25	99.45	99.42	99.76	99.68	99.39	99.27	99.53
	G-M	99.15	99.69	99.18	99.51	99.50	99.43	99.72	99.36	99.54	99.67
PART	Acc	98.83	98.86	98.49	99.88	98.48	99.65	98.38	99.88	98.83	98.29
	Sen	99.23	99.03	98.64	99.91	99.20	99.22	98.58	99.96	99.82	98.72
	Spe	99.79	98.87	99.17	99.54	98.81	99.93	99.08	99.59	99.52	99.17
	G-M	99.51	98.95	98.90	99.77	99.00	99.61	98.83	99.79	99.67	98.95
MLP	Acc	99.63	99.29	98.89	99.67	98.88	99.68	99.22	99.68	99.23	98.95
	Sen	99.33	99.11	99.77	99.37	98.90	99.19	99.18	99.85	99.21	99.47
	Spe	99.95	99.90	99.45	99.93	99.65	99.96	99.91	99.28	99.36	99.06
	G-M	99.66	99.55	99.61	99.68	99.27	99.59	99.59	99.56	99.28	99.26
RBFN	Acc	99.21	98.49	98.98	99.26	98.65	99.27	99.29	99.27	99.29	95.25
	Sen	99.43	99.42	99.45	99.18	98.69	99.55	99.32	99.34	99.31	95.83
	Spe	99.92	99.01	99.51	99.72	98.94	99.23	99.74	99.84	99.67	95.81
	G-M	99.71	99.22	99.48	99.45	98.81	99.39	99.52	99.59	99.49	95.82
SVM-P	Acc	99.65	99.68	99.27	99.63	99.28	99.68	98.92	99.62	98.97	98.69
	Sen	99.37	99.38	99.61	99.12	99.84	99.95	99.61	99.73	99.27	99.34
	Spe	99.11	99.19	99.23	99.73	99.60	99.21	99.21	99.40	99.22	99.39
	G-M	99.24	99.28	99.42	99.42	99.72	99.60	99.41	99.56	99.25	99.37
SVM-R	Acc	99.69	99.28	99.36	99.12	98.96	99.95	98.94	99.32	98.84	98.92
	Sen	99.67	99.40	99.38	99.28	99.84	99.37	99.31	99.35	99.65	99.33
	Spe	99.18	99.12	99.64	99.88	99.67	99.14	99.16	99.68	99.32	99.11
	G-M	99.42	99.26	99.51	99.58	99.76	99.25	99.24	99.51	99.49	99.22
LRA	Acc	98.86	98.89	97.53	99.38	97.96	98.98	98.62	99.36	97.92	98.36
	Sen	98.92	99.32	97.79	99.25	98.09	99.97	99.52	99.74	98.68	98.45
	Spe	99.08	99.43	97.72	99.82	98.20	99.13	99.01	99.93	98.10	99.15
	G-M	99.00	99.37	97.76	99.53	98.14	99.55	99.27	99.83	98.39	98.80
SMO	Acc	99.62	99.68	98.98	99.01	98.38	99.66	98.94	98.97	98.84	98.96
	Sen	99.86	99.12	99.11	99.53	99.23	99.82	99.92	99.89	99.65	99.41
	Spe	99.14	99.79	99.91	99.67	99.18	99.32	99.27	99.95	99.55	99.05
	G-M	99.50	99.45	99.51	99.60	99.21	99.57	99.60	99.92	99.60	99.23

are numerical attributes. The German-numerical dataset [99] contains 1000 instances with 24 attributes which describes the transactional history where 700 samples are approved applicants and 300 samples are rejected. For algorithms that need numerical attributes, Strathclyde University produced the file "German Data-numeric." Some attributes which are categorical have been converted as integer. Both the German (categorical and numerical) credit scoring datasets are loan approval datasets.

3.2 Performance measures

Various performance measures are available in the literature for evaluating the classification problems such as accuracy, sensitivity, specificity and G-measure. As most of the credit scoring approaches have considered, it is binary classification. And, accuracy measures the percentage of creditworthy and non-creditworthy cases are classified correctly. Mathematically, it is stated as follows in Eq. 3 [97]. Sensitivity and specificity also measure performance, but it takes into account only single class accuracy which alone is not sufficient, and are defined as in Eqs. 4 and 5, respectively. Another evaluation measure known as G-measure "as geometric mean of sensitivity and specificity" is as follows in Equation 6.

Table 9 Results on Bank-marketing dataset

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	CLFS	CFS
LDA	Acc	89.55	90.69	90.78	90.62	89.85	89.56	89.74	89.78	89.87	89.63
	Sen	80.60	81.62	81.70	81.56	80.87	80.60	80.77	80.80	80.88	80.67
	Spe	90.29	91.09	91.51	90.98	90.18	90.14	90.51	90.73	90.77	90.37
	G-M	85.31	86.23	86.47	86.14	85.39	85.24	85.50	85.62	85.68	85.38
NB	Acc	83.98	85.60	87.63	85.02	88.36	87.08	86.24	85.13	87.94	87.50
	Sen	76.51	77.49	79.19	76.91	80.19	78.45	78.43	77.57	79.70	78.91
	Spe	84.55	86.16	88.41	85.78	89.23	88.07	86.34	86.14	88.60	87.62
	G-M	80.43	81.71	83.67	81.22	84.59	83.12	82.29	81.74	84.03	83.15
TDNN	Acc	95.13	96.99	96.95	96.63	94.97	94.75	94.78	94.14	95.15	96.05
	Sen	87.21	88.86	89.06	88.51	87.36	87.15	87.21	85.84	87.57	88.06
	Spe	95.96	97.45	97.70	97.21	95.87	95.63	95.47	95.14	95.56	96.11
	G-M	91.48	93.06	93.28	92.76	91.51	91.29	91.24	90.37	91.48	92.00
KNN	Acc	87.98	88.03	87.72	88.76	87.76	88.01	88.56	88.43	87.98	88.32
	Sen	79.24	78.78	78.36	79.05	78.80	78.99	79.03	78.92	78.71	79.37
	Spe	88.70	88.15	87.94	89.43	88.74	88.28	89.10	89.34	88.48	88.36
	G-M	83.83	83.34	83.01	84.08	83.62	83.51	83.92	83.97	83.45	83.74
DT	Acc	86.68	86.92	86.79	87.41	84.53	86.90	88.60	88.85	86.98	87.32
	Sen	79.85	80.80	80.66	80.75	78.15	80.85	82.21	82.46	80.78	81.05
	Spe	86.87	87.12	86.84	88.28	85.04	87.05	89.02	88.90	87.31	87.49
	G-M	83.29	83.90	83.69	84.43	81.53	83.89	85.55	85.62	83.98	84.21
ELM	Acc	85.28	85.78	86.39	86.44	85.91	85.55	85.13	87.23	85.32	85.88
	Sen	79.24	78.94	80.25	79.69	79.42	79.64	79.26	80.73	78.63	79.47
	Spe	81.25	82.19	82.66	82.81	82.24	81.90	81.00	83.57	81.88	82.54
	G-M	80.24	80.55	81.44	81.24	80.82	80.76	80.13	82.14	80.23	80.99
ELM-1	Acc	87.59	87.87	88.47	88.56	88.54	87.79	89.22	89.27	87.61	88.63
	Sen	79.47	79.91	80.43	79.72	79.93	79.43	81.00	81.11	79.60	79.83
	Spe	80.77	80.92	82.06	81.67	82.30	81.36	82.40	82.42	81.54	81.57
	G-M	80.11	80.41	81.24	80.69	81.11	80.39	81.70	81.77	80.56	80.70
ELM-2	Acc	87.92	87.59	88.65	88.45	88.52	87.65	89.25	89.25	87.76	88.29
	Sen	79.99	78.94	79.87	79.83	80.41	79.25	81.26	80.65	79.68	80.04
	Spe	81.46	80.73	81.83	81.61	82.35	81.37	82.20	82.25	80.76	81.38
	G-M	80.72	79.83	80.85	80.71	81.37	80.30	81.73	81.44	80.22	80.71

$$\text{Accuracy (Acc)} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3}$$

$$\text{Sensitivity (Sen)} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4}$$

$$\text{Specificity (Spe)} = \frac{\text{TN}}{\text{TN} + \text{FN}} \tag{5}$$

$$\text{G} - \text{measure (G} - \text{M)} = \sqrt{\text{Sen} * \text{Spe}} \tag{6}$$

where Table 6 shows the descriptions about parameters involved to calculate the above performance measures.

4 Previous surveys on credit scoring models

This section presents various comparative analysis (or previous surveys) on credit scoring models toward the data mining and soft computing perspective, and detailed descriptions about surveys are as follows.

Lin et al. [65] have presented an article entitled “Machine Learning in Financial Crisis Prediction: A Survey.” This work focused on the development of state-of-the-art machine learning techniques, including hybrid and ensemble classifiers. In this article, authors have presented year-wise distribution of articles for ANNs, SVM, and GA; comparisons of single classifiers, related work using hybrid classification techniques; related work using ensemble tech-

Table 9 continued

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	CLFS	CFS
RF	Acc	89.06	89.42	88.98	89.78	88.38	89.07	88.38	88.54	89.07	88.89
	Sen	79.58	79.59	79.71	80.80	79.43	79.58	79.11	78.98	80.23	79.71
	Spe	83.10	83.33	83.53	84.40	83.13	83.48	83.18	83.11	82.87	82.96
	G-M	81.32	81.44	81.60	82.58	81.26	81.51	81.12	81.02	81.54	81.32
PART	Acc	88.29	89.40	89.07	89.01	88.69	88.47	89.09	88.91	88.56	88.56
	Sen	78.62	78.99	79.09	78.42	78.07	78.63	78.85	78.54	78.39	78.57
	Spe	79.58	80.51	80.63	80.23	80.34	79.70	81.08	80.92	79.82	80.62
	G-M	79.10	79.74	79.85	79.32	79.20	79.16	79.96	79.72	79.10	79.59
MLP	Acc	88.58	89.47	89.64	89.64	88.76	88.85	89.11	88.96	88.96	88.34
	Sen	80.76	82.19	82.51	82.53	80.89	81.10	81.41	81.69	81.26	81.03
	Spe	83.01	83.23	84.36	83.98	83.02	82.76	83.12	83.25	83.04	82.84
	G-M	81.88	82.71	83.43	83.25	81.95	81.92	82.26	82.46	82.14	81.93
RBFN	Acc	88.71	88.54	89.22	88.60	88.34	88.38	88.56	88.60	88.87	88.32
	Sen	79.75	79.48	80.20	79.57	78.78	79.17	79.19	79.28	79.18	79.36
	Spe	81.60	80.98	81.87	81.48	81.39	80.59	80.91	80.87	81.12	80.99
	G-M	80.67	80.23	81.03	80.52	80.07	79.88	80.05	80.07	80.14	80.17
SVM-P	Acc	87.12	87.36	87.86	87.82	87.43	87.82	88.61	88.37	87.82	87.38
	Sen	77.62	77.99	78.45	78.79	77.99	78.60	79.12	78.70	78.60	78.18
	Spe	80.77	81.15	81.72	81.35	81.25	81.08	82.13	82.10	81.20	80.63
	G-M	79.18	79.56	80.07	80.06	79.61	79.83	80.61	80.38	79.89	79.40
SVM-R	Acc	88.47	88.47	88.58	88.65	88.49	88.49	89.20	89.16	88.69	88.99
	Sen	77.87	78.07	78.59	78.41	78.12	78.32	78.76	78.77	78.78	78.83
	Spe	81.52	81.98	81.92	82.38	81.93	81.42	82.27	82.59	82.58	82.57
	G-M	79.68	80.00	80.24	80.37	80.00	79.85	80.50	80.66	80.66	80.68
LRA	Acc	89.86	89.89	89.93	89.91	90.12	88.96	89.16	89.18	89.42	90.19
	Sen	80.09	80.12	80.20	80.52	80.51	79.56	80.30	79.70	80.14	80.67
	Spe	83.66	84.37	84.59	83.67	84.28	83.19	83.21	83.47	83.71	83.92
	G-M	81.85	82.21	82.37	82.08	82.37	81.35	81.75	81.57	81.91	82.28
SMO	Acc	88.47	89.36	88.98	88.92	88.89	88.76	88.90	88.76	89.13	89.34
	Sen	79.14	80.28	79.68	79.82	79.57	79.59	79.62	79.44	79.90	80.04
	Spe	83.88	84.51	84.64	84.05	83.94	83.47	83.59	84.42	84.40	84.84
	G-M	81.48	82.37	82.12	81.91	81.72	81.51	81.58	81.89	82.12	82.41

niques; frequency of used baseline models and datasets. Lahsasna et al. [56] have presented a survey on state-of-the-art methods used in developing credit scoring models with issues discussed, especially from the data mining and soft computing point of view. Liang et al. [63] have applied various filter and wrapper-based feature selection methods on various diverse prediction models for the credit scoring data classification. Tsai [107] has presented comparative result analysis on feature selection methods such as T-test, correlation matrix, Stepwise, PCA and Factor Analysis (FA), and results are compared with the classification performance of MLP. From the experimental outcome, T-test is better to others and toward the reducing the number of features, Stepwise beats the others, which offers the lowest number of features from actual features, but the result of Stepwise for credit scoring problems shows unstable on the chosen five datasets. From the experimental results, T-test performs stably and provides higher prediction accuracy and results.

Abdou and Pointon [1] have presented an article entitled “credit scoring, statistical techniques and Evaluation criteria: a review of the literature.” In this study, authors have presented numerous performance assessment measures and different statistical techniques which are utilized especially in finance and banking enterprises. Moreover, a correlation between various factual methodologies exhibits that advanced strategies, for example, ANNs and genetic programming, perform superior to progressively customary systems, for example, DA and LR, as far as predictive

Table 10 Results on German-categorical dataset

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	CLFS	CFS
LDA	Acc	75.85	75.92	75.88	75.82	74.98	78.33	75.02	74.68	75.94	75.07
	Sen	69.58	69.51	69.68	69.76	68.27	72.24	68.28	68.73	69.29	68.66
	Spe	70.44	70.72	70.17	69.84	69.94	72.86	69.49	68.79	69.94	69.12
	G-M	70.01	70.11	69.92	69.80	69.10	72.55	68.88	68.76	69.61	68.89
NB	Acc	73.83	73.99	73.64	73.82	78.17	73.32	74.26	73.25	75.12	74.63
	Sen	65.65	65.77	65.48	65.10	69.03	65.17	66.15	65.45	66.40	66.05
	Spe	68.71	68.19	67.93	67.94	72.77	68.05	69.05	67.72	69.26	69.12
	G-M	67.16	66.97	66.69	66.50	70.88	66.59	67.58	66.58	67.81	67.57
TDNN	Acc	85.31	85.82	85.93	86.34	85.12	84.06	85.03	84.74	87.37	84.72
	Sen	76.90	77.29	77.84	78.14	77.24	75.75	76.85	76.42	79.06	76.59
	Spe	78.91	79.53	79.41	80.37	79.23	78.19	78.69	78.60	80.46	78.21
	G-M	77.90	78.40	78.62	79.25	78.23	76.96	77.77	77.50	79.76	77.39
KNN	Acc	66.49	68.38	70.49	68.29	68.47	66.51	69.53	70.58	66.81	70.63
	Sen	58.53	61.03	62.86	60.32	60.84	58.74	61.29	62.84	59.66	62.31
	Spe	60.34	62.40	64.24	61.91	61.87	60.20	63.46	63.93	60.32	63.64
	G-M	59.43	61.71	63.55	61.11	61.35	59.47	62.36	63.38	59.99	62.97
DT	Acc	74.97	75.96	74.35	76.36	74.84	73.86	73.98	77.43	77.86	75.64
	Sen	66.99	68.15	66.99	68.52	67.28	65.79	66.53	69.89	69.48	67.42
	Spe	69.18	70.46	69.38	70.30	69.85	68.63	68.34	71.38	72.16	70.00
	G-M	68.08	69.30	68.18	69.40	68.55	67.20	67.43	70.63	70.80	68.70
ELM	Acc	74.19	74.39	74.91	74.69	74.83	75.36	74.31	74.87	73.91	75.38
	Sen	67.45	67.43	67.82	67.81	68.23	68.02	67.61	67.83	67.42	68.57
	Spe	68.44	68.90	69.18	69.71	69.47	69.45	69.04	69.73	68.61	69.75
	G-M	67.94	68.16	68.50	68.75	68.85	68.73	68.32	68.77	68.01	69.16
ELM-1	Acc	74.56	75.62	75.36	75.89	75.88	76.22	75.66	75.79	74.83	76.73
	Sen	65.93	67.08	66.32	67.59	66.85	67.87	66.58	67.21	66.46	67.93
	Spe	66.48	67.99	67.61	67.79	68.02	68.38	67.62	67.63	67.55	69.00
	G-M	66.21	67.53	66.96	67.69	67.43	68.12	67.10	67.42	67.00	68.46
ELM-2	Acc	75.39	75.72	75.33	75.87	75.91	76.19	75.92	75.66	74.73	76.75
	Sen	67.99	69.01	68.12	68.72	69.09	68.60	69.23	68.67	68.12	69.30
	Spe	70.18	71.16	70.60	70.88	70.71	71.53	71.25	70.71	69.53	71.94
	G-M	69.08	70.08	69.35	69.79	69.89	70.05	70.23	69.68	68.82	70.61

performance. West [116] has applied various classification approaches such as MLP, mixture-of-experts, RBFN, LVQ, fuzzy adaptive resonance, etc. From the experimental results, it is found that LR is the most accurate approach for credit scoring. Desai et al. [18] have investigated the predictive power of feed-forward neural networks with back propagation in comparison with traditional techniques such as LDA and LR. A specific preferred standpoint offered by ANNs is that a pre-determination of the model isn't required since subsidizing and little example estimate regularly block the utilization of modified credit scoring models at small credit associations. The performance of generic models as contrasted and customized ANNs offers a promising road if the proportion of execution is level of terrible advances effectively arranged. In any case, if the measure of performance is percentage of good and bad credits accurately characterized, LR models are practically more identical to the ANNs approach. In this study, Yeh and Lien [124] studies six classification approaches, namely KNN, NB, ANN, Discriminate analysis (DA), LR, Classification Tree. From the experimental outcomes, ANN performs classification more accurately than the other five methods.

Abellán and Castellano [2] have offered a comparative study on numerous ensemble methods for credit scoring. Standard credit scoring models rely upon individual classifiers or an essential blend of these classifiers which will by and large show moderate execution. Tripathi et al. [101] have presented a comparative result analysis on STEP-, CART-,

Table 10 continued

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	CLFS	CFS
RF	Acc	77.29	77.48	77.91	78.38	78.87	78.64	78.49	77.72	78.95	78.06
	Sen	69.86	69.94	70.60	71.23	71.42	70.83	70.73	70.18	71.74	70.82
	Spe	72.32	72.15	72.84	73.82	73.60	73.49	73.12	72.44	74.30	73.35
	G-M	71.08	71.04	71.71	72.51	72.50	72.15	71.91	71.30	73.01	72.07
PART	Acc	71.67	70.36	69.28	70.79	70.97	71.59	71.96	71.90	72.83	72.36
	Sen	63.81	62.87	61.87	63.18	63.91	64.12	64.36	64.91	65.61	64.69
	Spe	64.71	63.57	63.17	63.79	64.61	65.00	65.05	64.80	65.56	65.34
	G-M	64.25	63.22	62.52	63.48	64.26	64.56	64.71	64.85	65.59	65.01
MLP	Acc	74.14	75.18	74.39	76.02	75.36	75.92	75.16	73.97	75.33	75.39
	Sen	67.09	68.15	67.42	69.17	68.82	69.21	68.30	67.31	68.37	68.58
	Spe	69.56	70.68	69.64	71.23	71.01	70.63	70.48	69.72	70.77	70.74
	G-M	68.32	69.40	68.52	70.20	69.91	69.92	69.38	68.51	69.56	69.65
RBFN	Acc	74.36	75.03	74.46	73.86	74.62	73.84	74.39	74.82	74.87	74.93
	Sen	68.05	69.06	68.73	67.30	67.96	67.58	68.03	68.81	69.09	69.18
	Spe	69.79	70.51	70.16	69.29	69.85	69.18	69.74	70.04	70.25	70.55
	G-M	68.91	69.78	69.44	68.29	68.90	68.38	68.88	69.42	69.67	69.86
SVM-P	Acc	74.28	74.96	76.98	74.58	75.39	74.83	74.93	75.74	74.04	75.31
	Sen	67.07	67.29	69.12	67.29	67.17	67.29	67.12	67.81	66.06	67.89
	Spe	69.91	70.47	71.88	69.83	70.86	70.17	70.16	71.22	69.18	70.42
	G-M	68.48	68.86	70.49	68.55	68.99	68.72	68.62	69.50	67.60	69.14
SVM-R	Acc	75.82	75.93	76.78	75.73	75.91	75.21	75.26	76.21	75.53	75.99
	Sen	69.34	69.69	69.93	69.77	69.73	68.85	68.71	70.07	69.49	69.66
	Spe	70.23	70.86	71.33	70.33	70.54	69.54	69.95	70.65	70.43	70.23
	G-M	69.78	70.27	70.63	70.05	70.14	69.20	69.33	70.36	69.96	69.95
LRA	Acc	75.91	74.61	73.07	74.91	77.51	74.13	72.37	75.43	77.57	75.91
	Sen	67.51	65.97	65.00	66.07	68.73	65.37	63.87	67.14	68.67	67.39
	Spe	68.61	67.99	65.83	67.63	70.27	67.27	65.47	68.71	69.94	69.02
	G-M	68.05	66.97	65.41	66.85	69.50	66.32	64.66	67.92	69.30	68.20
SMO	Acc	75.42	74.93	74.76	76.38	73.92	75.69	74.81	74.83	76.90	76.34
	Sen	68.45	67.91	67.66	69.03	67.05	68.27	67.74	67.94	70.05	69.53
	Spe	70.61	69.73	70.47	71.28	69.40	70.81	69.58	70.36	71.60	71.29
	G-M	69.52	68.81	69.05	70.14	68.21	69.53	68.65	69.14	70.82	70.41

CORR-, MARS-, T-test-, and NRS-based feature selection with QDA-, MLFN-, DT-, TDNN-, DTNN-, PNN-, and NB-based feature selection. From the results, it is observed that STEP with TDNN has the best classification performance and it also improves the classification performances as compared to performance with all features. Along with that, this paper has also presented the comparative analysis on ensemble frameworks with all aforementioned feature selection approaches. From the results, it is observed that STEP with layered ensemble framework with weighted voting as aggregation function has the best classification performance. Nanni and Lumini [73] have presented a comparative analysis on ensemble classifiers for bankruptcy prediction; from the experimental observation it is concluded that random subspace with Levenberg–Marquardt Neural Nets (LMNC) has the best performance as compared to rest of the approaches.

Wang et al. [113] have applied a number of ensemble learning methods with LRA, DT, ANN and SVM. From the exploratory outcomes, it uncovers that the utilization of ensemble learning has been brought significant improvement for individual base classifier. Bagging performs superior to Boosting. Also, Stacking and Bagging DT get best outcomes as far as outcomes. Also, among four base learners DT gets best improvement as far as accuracy with ensemble learning.

In this study in Sect. 2, we have presented a comprehensive literature survey on various existing approaches applied for credit scoring data analysis. In this section, we have presented various surveys on credit scoring and found that these

Table 11 Results on German-numerical dataset

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	CLFS	CFS
LDA	Acc	76.60	76.67	76.63	76.57	75.72	79.11	75.77	75.42	76.69	75.82
	Sen	67.48	68.37	67.92	67.51	67.44	70.26	66.88	67.23	68.02	66.86
	Spe	69.76	70.60	70.11	69.94	69.56	72.82	69.56	69.30	70.49	69.48
	G-M	68.61	69.48	69.00	68.71	68.49	71.53	68.21	68.26	69.25	68.16
NB	Acc	74.56	74.72	74.37	74.55	78.95	74.05	75.02	73.98	75.87	75.37
	Sen	66.61	66.66	65.60	65.69	70.26	65.67	66.95	65.93	66.89	66.46
	Spe	67.13	67.11	66.64	66.54	70.60	66.33	67.17	66.34	67.73	67.30
	G-M	66.87	66.88	66.12	66.11	70.43	66.00	67.06	66.14	67.31	66.88
TDNN	Acc	86.16	86.67	86.78	87.20	85.97	84.90	85.88	85.58	88.24	85.56
	Sen	76.80	77.97	77.65	78.41	77.18	76.26	77.05	76.75	78.60	76.50
	Spe	80.55	80.98	81.45	81.43	80.34	79.58	79.90	80.15	82.11	79.92
	G-M	78.66	79.46	79.53	79.91	78.75	77.90	78.46	78.43	80.34	78.19
KNN	Acc	67.15	69.06	71.19	68.97	69.15	67.17	70.22	71.28	67.47	71.33
	Sen	59.35	61.17	63.42	61.44	61.08	59.86	62.19	63.64	59.47	63.28
	Spe	60.45	62.44	64.34	63.01	62.39	60.82	63.65	64.81	61.59	64.20
	G-M	59.90	61.80	63.88	62.22	61.73	60.34	62.92	64.22	60.52	63.74
DT	Acc	75.71	76.71	75.09	77.12	75.58	74.59	74.71	78.20	78.63	76.39
	Sen	68.28	68.40	67.45	69.13	67.42	67.02	66.80	69.92	70.56	68.71
	Spe	72.00	72.27	71.50	72.78	71.45	70.32	71.08	74.37	74.37	72.02
	G-M	70.12	70.31	69.44	70.93	69.40	68.65	68.91	72.11	72.44	70.35
ELM	Acc	74.93	75.13	75.65	75.43	75.57	76.11	75.05	75.61	74.64	76.13
	Sen	68.98	68.83	69.34	69.04	69.63	70.16	68.80	69.17	68.59	69.32
	Spe	71.62	71.55	72.27	72.15	71.98	72.84	72.29	72.58	71.11	72.59
	G-M	70.29	70.18	70.79	70.58	70.79	71.49	70.53	70.85	69.84	70.94
ELM-1	Acc	75.30	76.37	76.11	76.64	76.63	76.98	76.41	76.54	75.57	77.49
	Sen	69.49	70.35	70.12	70.62	70.55	71.15	70.58	71.32	70.28	71.87
	Spe	72.57	73.33	73.99	74.03	74.31	74.83	74.30	74.27	73.13	74.78
	G-M	71.01	71.83	72.03	72.31	72.40	72.96	72.42	72.78	71.69	73.31
ELM-2	Acc	76.14	76.47	76.08	76.62	76.66	76.95	76.67	76.41	75.47	77.51
	Sen	71.66	72.00	71.51	71.83	71.71	71.66	72.14	71.89	71.08	72.23
	Spe	73.84	74.03	73.81	73.71	74.29	74.02	73.90	73.52	73.12	75.32
	G-M	72.74	73.01	72.65	72.76	72.99	72.83	73.01	72.70	72.09	73.76

surveys presents the surveys or comparative analysis on some machine learning approaches. The existing approaches have good performances; however, more classification and feature selection approaches are available in literature which are not applied in case of credit scoring dataset. Without applying specific approach with specific dataset, it is unpredictable to analyze its performance. Hence, this study present comparative analysis of various state-of-the-art approaches for feature selection and classification on credit scoring datasets and results are presented in terms of classification accuracy, sensitivity, specificity and G-measure. So, toward comparative result analysis, we have considered nine feature selection approaches which are not yet applied in this domain with sixteen heterogeneous classifiers such as LDA-, NB-, KNN-, ANNs-, SVMs- and tree-based classifiers. From the literature, we have observed that most of researches have considered few credit scoring dataset and presented a comparative analysis with few classifiers. So, in this study, we have considered seven different datasets related to credit scoring such as credit card application datasets, loan approval datasets, bank marketing dataset and existing credit card customers dataset. As, main motivation of this study is to provide a experimental result analysis of credit scoring models with machine learning approaches.

Table 11 continued

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	CLFS	CFS
RF	Acc	78.06	78.25	78.68	79.16	79.65	79.42	79.27	78.49	79.73	78.84
	Sen	72.96	73.05	73.35	74.08	74.33	74.15	74.05	73.68	74.82	73.55
	Spe	75.86	75.13	75.70	76.27	76.82	76.82	76.92	76.16	77.18	76.47
	G-M	74.40	74.08	74.51	75.17	75.57	75.47	75.47	74.91	75.99	74.99
PART	Acc	72.38	71.06	69.97	71.49	71.67	72.30	72.67	72.61	73.55	73.08
	Sen	64.22	62.62	62.04	62.94	63.48	63.79	64.17	64.87	65.60	64.59
	Spe	65.74	64.42	63.78	64.81	65.31	65.18	66.32	66.19	66.88	66.14
	G-M	64.98	63.51	62.90	63.87	64.39	64.48	65.24	65.53	66.23	65.36
MLP	Acc	74.88	75.93	75.13	76.78	76.11	76.67	75.91	74.70	76.08	76.14
	Sen	67.96	68.52	68.47	69.22	68.53	69.31	68.64	67.42	69.18	69.18
	Spe	69.81	70.10	69.76	70.89	70.49	71.45	70.10	69.31	70.71	70.05
	G-M	68.88	69.30	69.11	70.05	69.50	70.37	69.36	68.36	69.94	69.61
RBFN	Acc	75.10	75.78	75.20	74.59	75.36	74.57	75.13	75.56	75.61	75.67
	Sen	68.41	68.31	67.89	67.56	68.04	67.60	67.76	68.57	68.48	68.11
	Spe	70.37	71.46	70.53	69.89	70.65	69.92	70.68	70.79	71.28	70.38
	G-M	69.38	69.87	69.20	68.71	69.33	68.75	69.21	69.67	69.86	69.24
SVM-P	Acc	75.02	75.70	77.74	75.32	76.14	75.57	75.67	76.49	74.78	76.06
	Sen	67.98	68.62	70.45	68.05	68.89	68.83	68.42	69.30	67.43	68.85
	Spe	71.31	71.30	73.99	71.23	72.15	71.66	71.91	72.00	71.25	72.49
	G-M	69.63	69.95	72.20	69.62	70.50	70.23	70.14	70.64	69.31	70.65
SVM-R	Acc	76.57	76.68	77.54	76.48	76.66	75.96	76.01	76.97	76.28	76.74
	Sen	70.55	70.91	71.80	70.94	71.49	70.72	69.98	71.74	70.59	71.38
	Spe	72.91	73.78	73.69	72.71	73.37	73.14	72.53	73.34	72.48	73.44
	G-M	71.72	72.33	72.74	71.82	72.42	71.92	71.25	72.54	71.53	72.40
LRA	Acc	76.66	75.35	73.80	75.65	78.28	74.87	73.09	76.18	78.34	76.66
	Sen	68.29	66.51	65.53	67.04	69.56	66.14	64.60	67.20	69.71	67.84
	Spe	70.35	68.94	67.58	69.17	71.65	68.33	67.38	70.19	71.51	70.73
	G-M	69.31	67.71	66.55	68.10	70.59	67.23	65.98	68.68	70.61	69.27
SMO	Acc	76.17	75.67	75.50	77.14	74.65	76.44	75.55	75.57	77.66	77.10
	Sen	69.29	68.80	68.74	69.71	68.07	68.80	68.97	68.34	70.29	69.64
	Spe	73.32	72.82	72.94	74.69	71.70	73.80	73.03	73.51	74.58	74.40
	G-M	71.27	70.78	70.80	72.16	69.86	71.26	70.97	70.88	72.41	71.98

5 Results and discussion

In this article, we have applied nine feature selection approaches such as ILFS, ECFS, Relief, FSV, LS, MFCS, UDFS, LLCFS and CFS with various classification approaches such as LDA, NB, TDNN, KNN, DT, ELM, ELM-1, ELM-2, RF, PART, MLP, RBFN, SVM (with RBF and Poly kernels as SVM-R and SVM-P, respectively), LRA and SMO. Tenfold Cross-Validation is applied, 50 iterations are applied, and mean is considered for the comparative results analysis with seven aforementioned credit scoring dataset. Before applying the feature selection approach, some pre-processing steps such as data cleaning and data transformation are applied. These datasets have some missing values and have both numerical along with categorical attributes. Some classification such as neural network-based classification approach can't work with categorical datasets. So, toward data cleaning, samples with missing values are eliminated, and toward data transformation each categorical value in each feature is replaced by a unique integer value. Further pre-processed dataset is considered for result analysis; classification performances of sixteen classifiers with nine approaches for feature selection along with actual set of features (as All) on seven credit scoring datasets are tabulated in Tables 7, 8, 9, 10, 11, 12 and 13.

Starting form result analysis on Australian dataset as tabulated in Table 7 with respective feature selection and classification approaches, in this dataset, TDNN has the best and

Table 12 Results on Japanese dataset

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	CLFS	CFS
LDA	Acc	86.62	86.73	86.27	87.20	87.79	87.22	87.98	87.39	86.23	88.39
	Sen	85.29	85.56	84.85	85.71	86.19	86.00	86.23	86.01	85.03	86.94
	Spe	87.14	92.70	92.16	90.47	89.08	93.08	89.32	87.58	88.82	90.07
	G-M	86.21	89.06	88.43	88.06	87.62	89.47	87.76	86.79	86.90	88.49
NB	Acc	85.04	86.07	85.82	87.49	86.51	85.93	85.77	85.88	85.64	85.99
	Sen	84.31	84.12	84.92	89.01	88.85	85.31	87.18	87.25	83.93	84.17
	Spe	86.03	86.63	87.10	85.92	85.79	87.17	84.07	85.18	86.78	87.32
	G-M	85.17	85.37	86.00	87.45	87.31	86.23	85.61	86.21	85.34	85.73
TDNN	Acc	88.92	89.89	88.99	89.02	90.39	88.18	88.45	88.22	90.44	89.58
	Sen	89.87	89.91	89.87	89.76	92.15	89.21	87.34	89.56	92.05	89.64
	Spe	88.56	88.86	88.89	89.94	89.46	88.73	89.86	88.42	89.12	91.57
	G-M	89.21	89.38	89.38	89.85	90.80	88.97	88.59	88.99	90.57	90.60
KNN	Acc	75.66	74.62	73.83	74.01	76.34	74.15	79.02	74.36	74.17	75.12
	Sen	72.79	76.69	76.22	78.67	82.55	75.89	72.51	77.03	79.98	89.97
	Spe	75.76	74.88	74.63	75.51	76.42	75.40	79.77	75.85	74.99	77.19
	G-M	74.26	75.78	75.42	77.07	79.42	75.64	76.05	76.43	77.44	83.34
DT	Acc	83.17	82.81	382.24	83.64	81.55	80.84	81.65	84.81	82.98	83.64
	Sen	85.49	84.36	84.23	85.22	84.30	81.10	84.64	87.69	83.15	83.85
	Spe	83.47	83.61	82.48	84.62	82.10	80.84	81.87	85.26	83.61	84.49
	G-M	84.48	83.98	83.35	84.92	83.20	80.97	83.25	86.46	83.38	84.17
ELM	Acc	81.36	81.01	81.38	82.79	81.81	81.78	82.29	83.98	82.97	83.88
	Sen	82.16	82.00	81.84	83.54	82.30	81.82	83.21	84.49	83.05	84.12
	Spe	81.86	82.53	84.00	83.00	84.20	84.82	85.58	84.71	83.22	85.77
	G-M	82.01	82.26	82.92	83.27	83.25	83.31	84.39	84.60	83.14	84.94
ELM-1	Acc	81.83	82.84	83.33	83.00	82.02	82.73	85.98	86.15	82.85	83.37
	Sen	82.74	83.31	83.82	83.83	82.61	81.61	86.63	86.63	83.40	83.53
	Spe	85.60	85.91	86.90	86.13	83.93	84.23	89.56	89.98	83.87	87.26
	G-M	84.16	84.60	85.35	84.97	83.26	82.91	88.08	88.29	83.63	85.38
ELM-2	Acc	83.83	83.92	83.93	83.52	80.76	83.72	85.98	86.15	83.07	84.09
	Sen	83.97	84.09	83.94	83.62	81.09	84.31	86.79	86.27	83.96	84.52
	Spe	83.85	84.12	84.36	84.00	81.44	84.32	86.57	86.86	83.63	84.94
	G-M	83.91	84.10	84.15	83.81	81.27	84.32	86.68	86.57	83.79	84.73

RF has the second best classification accuracies. LDA and LRA with CFS, NB with LFCS, TDNN, RF with ILFS, KNN with ILFS, DT, ELM, PART, MLP, SVM-R and SMO with UDFS, ELM-1, ELM-2, RBFN, SVM-P with MFCS are best combinations, and these combinations make the significant improvement on classification performance. Overall conclusion on Australian dataset, UDFS has best performances with all most aforementioned classifiers and it also improves the classification performances as compared to performances with all features. From comparing the results as shown in Table 7 and results in Tables 1, 2, 3, and 4 (results from prior works), TDNN with CFS has the best classification performances except than ensemble classification approaches such as NRS + LWV and LWV.

Next, result analysis on Bankruptcy dataset is tabulated in Table 8 with respective feature selection and classification approaches. In this dataset, ELM-1 has the best which is closed to TDNN, ELM, RF, SVM and SMO classification accuracies. And, aforementioned feature selection approaches have the similar impact with all aforementioned classification approaches, because this dataset has less number of features & samples and separability of data samples.

Result analysis on Bank marketing dataset is tabulated in Table 9 with respective feature selection and classification approaches. In this dataset, TDNN has the best and LRA & LDA have the second best classification accuracies. LDA with ECFS has the best classification performances which is closed to ILFS- and Relief-based feature selection's per-

Table 12 continued

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	CLFS	CFS
RF	Acc	88.12	85.94	88.16	89.23	89.81	88.34	87.78	87.65	88.42	89.23
	Sen	86.92	86.26	88.25	89.24	89.99	89.13	88.26	88.21	89.32	90.12
	Spe	88.89	85.99	88.82	90.05	90.41	88.87	87.94	88.64	88.62	89.85
	G-M	87.91	86.13	88.53	89.65	90.20	89.00	88.10	88.43	88.97	89.99
PART	Acc	83.91	84.35	82.74	84.49	85.22	81.76	82.76	84.93	83.20	84.49
	Sen	84.88	84.38	83.12	85.08	86.01	82.54	83.38	85.07	84.12	85.42
	Spe	84.74	85.00	83.03	85.30	85.32	82.71	83.16	85.25	83.97	85.08
	G-M	84.81	84.69	83.07	85.19	85.66	82.62	83.27	85.16	84.05	85.25
MLP	Acc	83.77	83.91	83.06	84.64	84.35	83.20	84.62	85.94	82.96	86.52
	Sen	84.18	84.15	83.75	85.25	84.83	84.17	85.54	85.99	83.90	86.84
	Spe	84.55	84.08	83.65	88.52	85.22	87.34	88.70	89.41	83.19	89.05
	G-M	84.36	84.11	83.70	86.87	85.03	85.74	87.11	87.68	83.55	87.94
RBFN	Acc	82.75	83.33	82.32	84.64	82.46	82.20	84.62	84.40	82.75	84.91
	Sen	83.45	83.92	82.62	84.68	83.30	82.27	84.67	84.83	83.01	85.37
	Spe	83.50	83.58	82.40	85.01	83.07	83.08	85.55	85.36	82.97	85.76
	G-M	83.47	83.75	82.51	84.85	83.19	82.67	85.11	85.09	82.99	85.56
SVM-P	Acc	82.49	81.59	82.36	83.26	83.86	82.30	82.29	82.41	83.87	83.78
	Sen	82.84	82.43	83.30	84.21	84.11	82.93	83.19	82.46	84.75	84.05
	Spe	83.25	81.76	82.88	84.17	84.71	83.30	82.98	82.71	84.66	84.69
	G-M	83.05	82.10	83.09	84.19	84.41	83.11	83.09	82.59	84.71	84.37
SVM-R	Acc	83.79	83.73	83.30	84.49	84.09	84.96	85.93	85.64	83.51	83.96
	Sen	83.89	84.07	84.22	85.44	85.07	85.28	86.82	86.18	83.75	84.30
	Spe	84.37	83.84	84.26	84.75	84.80	85.55	86.03	85.79	83.99	84.01
	G-M	84.13	83.95	84.24	85.10	84.93	85.42	86.42	85.99	83.87	84.16
LRA	Acc	85.94	85.51	84.66	85.65	85.80	85.82	85.65	85.94	85.07	85.94
	Sen	84.87	85.57	85.44	85.98	86.44	86.58	86.42	86.61	85.55	86.10
	Spe	87.23	85.82	85.65	86.47	86.66	86.69	86.03	86.52	85.65	86.37
	G-M	86.04	85.69	85.55	86.22	86.55	86.64	86.23	86.56	85.60	86.24
SMO	Acc	85.65	85.99	84.62	86.79	86.36	84.87	86.78	86.66	85.56	87.06
	Sen	89.05	88.50	86.84	88.22	89.47	86.33	89.34	87.06	85.81	91.04
	Spe	85.64	85.93	85.39	86.73	88.12	87.08	86.16	85.90	85.00	88.89
	G-M	87.33	87.21	86.11	87.47	88.79	86.70	87.73	86.48	85.40	89.96

formances. NB with FSV has the best performances, TDNN with ECFS has the best classification performances which is closed to ILFS-, UDFS- and Relief-based feature selection's performances, and KNN with Relief has the best classification performances which is closed to FSV- and MCFS-based feature selection's performances. DT with MCFS has the best and is closed to UDFS-based feature selection's performances. ELM with UDFS has the best performances, ELM-1 & ELM-2 with MFCS have the best and is closed to UDFS-based feature selection's performances. RF with Relief, PART with UDFS, RBFN with ECFS, SVM-P and SVM-R with MFCS are the best combinations, and these sets make the significant improvement on classification performances. MLP with Relief has the best performances and is closed to ECFS performances. LRA with CFS has the best performances and is closed to FSV performances. SMO with ILFS has the best performances and is closed to CFS performances. Overall conclusion on Bank-marketing datasets, UDFS and MFCS have the best performances with all most aforementioned classifiers and it also improves the classification performance as compared to performance with all features.

Further, result analysis on German-categorical and numerical datasets is tabulated in Tables 10 and 11 with respective feature selection and classification approaches. In these datasets, TDNN has the best and LRA, SVM-R & LDA have the second best classification accuracies. LDA with LS; NB with FSV; TDNN, DT, LRA, RF & PART with LLCFS;

Table 13 Results on Taiwan dataset

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	LLCFS	CFS
LDA	Acc	81.06	81.11	81.13	81.29	81.65	81.88	81.77	80.32	80.65	83.13
	Sen	70.63	69.88	69.89	70.62	71.14	71.20	70.54	69.51	70.34	72.30
	Spe	74.63	74.86	75.11	75.60	75.77	76.01	75.91	74.09	74.57	77.13
	G-M	72.60	72.33	72.46	73.07	73.42	73.57	73.18	71.76	72.42	74.68
NB	Acc	71.05	75.12	77.13	75.12	70.98	70.61	75.14	76.36	71.98	77.13
	Sen	65.49	68.83	70.26	68.62	65.01	64.57	68.60	69.59	65.84	70.72
	Spe	68.12	71.79	73.30	71.59	68.29	67.87	71.47	72.82	69.03	73.42
	G-M	66.79	70.29	71.77	70.09	66.63	66.20	70.02	71.19	67.42	72.06
TDNN	Acc	81.63	82.62	81.81	81.65	81.60	80.82	82.66	81.06	81.60	82.81
	Sen	74.52	75.20	74.51	75.19	75.20	73.94	75.44	74.49	74.82	75.96
	Spe	79.05	79.88	79.09	79.16	79.15	77.70	80.31	78.64	78.99	80.38
	G-M	76.75	77.50	76.77	77.15	77.15	75.79	77.84	76.54	76.87	78.14
KNN	Acc	89.02	89.01	89.44	88.77	87.89	88.08	89.15	88.55	87.89	89.44
	Sen	78.43	79.20	79.32	78.86	77.81	78.35	78.98	78.41	77.49	79.38
	Spe	81.04	81.19	81.79	80.83	80.71	80.48	81.45	81.13	80.27	82.37
	G-M	79.73	80.19	80.55	79.84	79.25	79.40	80.20	79.76	78.87	80.86
DT	Acc	70.95	70.09	71.87	71.79	70.80	71.10	69.11	71.16	71.80	71.87
	Sen	63.40	63.29	64.07	64.82	63.56	63.83	62.37	64.16	64.70	63.98
	Spe	64.73	64.40	66.09	66.29	64.85	65.19	63.36	65.15	66.25	65.69
	G-M	64.06	63.84	65.07	65.55	64.20	64.51	62.86	64.65	65.47	64.83
ELM	Acc	77.42	77.88	78.39	78.22	78.18	76.42	78.28	76.63	78.18	77.39
	Sen	72.00	71.74	72.15	72.06	72.72	71.04	72.99	70.51	72.11	71.93
	Spe	73.71	74.86	74.49	74.59	75.19	72.65	74.41	73.68	75.21	73.66
	G-M	72.85	73.28	73.31	73.31	73.94	71.84	73.70	72.08	73.65	72.79
ELM-1	Acc	79.13	79.61	78.87	79.81	79.89	80.12	81.84	79.50	80.89	80.27
	Sen	74.09	74.38	74.03	74.33	75.30	74.98	76.23	74.53	76.16	74.88
	Spe	75.53	76.47	75.86	76.15	76.82	76.17	78.25	76.41	76.87	76.71
	G-M	74.81	75.42	74.94	75.23	76.06	75.57	77.24	75.46	76.51	75.79
ELM-2	Acc	80.89	80.86	80.15	80.92	79.59	79.34	83.69	83.25	80.34	82.88
	Sen	75.58	76.05	74.87	76.03	74.54	74.08	78.58	78.26	75.59	77.75
	Spe	79.96	79.77	79.41	79.44	78.51	78.11	82.32	81.90	78.74	82.12
	G-M	77.74	77.89	77.10	77.72	76.50	76.07	80.43	80.06	77.15	79.90

ELM, RBFN & MLP with CFS; SVM-P & SVM-R with ECFS have the best classification performances. KNN with CFS has the best performances in case of German-categorical dataset, and it with UDFS which is closed to ECFS has the best performances in case of German-numerical dataset. ELM-1 & ELM-2 with CFS and MFCS have the best performances in German categorical and numerical datasets, respectively. SMO has the best performances with Relief-based feature selection and is closed to CFS-based feature selection with both datasets. Overall conclusion on German-categorical/numerical datasets, CFS and LLCFS have best performances with all most aforementioned classifiers and it also improves the classification performances as compared to performances with all features. From comparing the results as shown in Table 10 and results in Tables 1, 2, 3, and 4 (results obtained from prior works), TDNN with LLCFS has the best classification performances except than ensemble classification approaches such as NRS + LWV and FS + WV.

In case of Japanese dataset, results as are tabulated in Table 12 with respective feature selection and classification approaches. In this dataset, TDNN has the best and RF has the second best classification accuracy. LDA, NB, MLP, RBFN, LRA and SMO with CFS, RF with ILFS, PART with FSV have the best performances; TDNN with LLCFS has the best performances and is closed to FSV. KNN, ELM-1 & SVM-R with MFCS, DT & ELM-2 with UDFS are the best combinations. ELM with UDFS has the best performances and is closed to CFS. SVM-P with LLCFS has the best perfor-

Table 13 continued

	Measures	All	ILFS	ECFS	Relief	FSV	LS	MFCS	UDFS	LLCFS	CFS
RF	Acc	77.15	78.34	77.12	78.81	78.96	78.19	77.86	76.35	78.96	78.12
	Sen	72.72	73.32	71.97	73.54	74.18	73.06	72.75	71.87	74.16	73.65
	Spe	74.72	76.16	74.70	76.04	76.50	75.50	75.31	73.45	76.55	75.16
	G-M	73.71	74.72	73.33	74.78	75.33	74.27	74.02	72.66	75.35	74.40
PART	Acc	79.77	80.18	80.41	80.54	79.12	79.36	78.44	79.34	80.12	80.37
	Sen	70.65	71.55	71.36	71.57	70.04	70.01	69.08	70.06	70.87	71.36
	Spe	73.52	74.00	74.02	74.22	73.56	73.24	73.06	73.31	73.77	74.86
	G-M	72.07	72.76	72.68	72.88	71.78	71.61	71.04	71.66	72.31	73.09
MLP	Acc	81.43	82.84	82.94	82.68	80.91	80.87	82.44	81.62	81.91	83.44
	Sen	76.24	77.71	77.31	77.04	76.15	76.20	77.29	76.20	76.52	77.72
	Spe	78.57	79.70	79.65	79.38	77.89	77.64	79.44	78.64	78.70	80.57
	G-M	77.40	78.70	78.47	78.20	77.01	76.92	78.36	77.41	77.60	79.13
RBFN	Acc	81.68	83.82	82.84	82.78	82.64	81.29	82.92	81.84	82.64	83.64
	Sen	75.06	77.06	75.55	75.91	75.63	74.64	76.21	74.61	75.62	76.53
	Spe	77.39	79.58	78.86	77.88	78.15	76.92	78.72	77.04	78.51	79.60
	G-M	76.22	78.31	77.19	76.89	76.88	75.77	77.46	75.82	77.05	78.05
SVM-P	Acc	79.10	79.13	80.05	81.75	80.65	77.88	80.74	79.93	78.65	81.74
	Sen	74.48	74.10	75.04	76.54	75.98	72.98	75.60	74.42	73.16	76.28
	Spe	73.16	73.51	74.10	75.23	74.39	71.83	75.18	73.77	72.99	75.99
	G-M	73.82	73.80	74.57	75.88	75.18	72.40	75.39	74.09	73.08	76.14
SVM-R	Acc	78.97	79.93	78.28	79.94	80.89	80.09	80.14	80.23	80.89	80.92
	Sen	73.43	74.35	72.32	74.09	75.08	74.37	74.02	74.79	74.73	75.04
	Spe	73.60	74.85	73.02	74.78	75.41	75.28	74.95	75.22	75.79	75.70
	G-M	73.51	74.60	72.67	74.43	75.25	74.83	74.48	75.01	75.25	75.37
LRA	Acc	79.63	79.87	80.08	80.25	80.73	79.93	80.09	80.36	80.73	81.17
	Sen	75.03	74.74	74.85	75.63	75.78	74.56	75.18	75.44	75.67	76.09
	Spe	73.54	73.93	74.06	74.63	75.16	74.00	74.57	74.30	74.75	75.13
	G-M	74.28	74.33	74.45	75.13	75.47	74.28	74.88	74.87	75.20	75.61
SMO	Acc	81.03	81.82	81.59	81.71	80.64	81.87	81.92	80.21	81.64	83.22
	Sen	73.85	74.14	74.21	73.98	73.35	74.66	74.64	72.48	74.07	75.85
	Spe	75.65	76.74	76.87	76.47	75.13	76.54	76.31	74.64	76.51	77.40
	G-M	74.74	75.43	75.53	75.21	74.24	75.60	75.47	73.55	75.28	76.62

mances and is closed to CFS and FSV. Overall conclusion on Japanese dataset, CFS has the best performances with all most aforementioned classifiers and it also improves the classification performance as compared to performances with all features. From comparing the results as shown in Table 12 and results in Tables 1, 2, 3, and 4 (results obtained from prior works), TDNN with CFS has the best classification performances.

At last with Taiwan dataset, the result analysis from the results is tabulated in Table 13 with respective feature selection and classification approaches. In this dataset, TDNN has the best and MLP & RBFN has the second best classification accuracies and G-measures. LDA, NB, TDNN, KNN, DT, MLP, RBFN, SVM-R, LRA and SMO with CFS, ELM with ECFS, ELM-1 & ELM-2 with MCFS, RF, PART & SVM-P with Relief have the best performances. Overall conclusion on Taiwan dataset, CFS has best performances with all most aforementioned classifiers and it also improves the classification performance as compared to performances with all features.

Overall, as in this study, various classification approaches applied with various feature selection approaches and with all features on various credit scoring datasets and results are compared in terms of classification performances. Further, results obtained from literature are also compared. From the experimental outcomes of this study, it can be concluded as TDNN and RF are the best and second performers with most of the aforementioned credit scoring datasets. Usually, all aforementioned feature selection approaches have made significant improvements toward classification accuracy, but CFS has the best values with most of the classification approaches. Further, by comparing the results of this study with same of from prior works, we found best values as compared to classification and hybrid approaches except than ensemble classification approaches such as NRS + LWV and FS + WV (as this study focused on hybrid approaches). Further, by applying the ensemble classification approaches with consideration of feature selection and TDNN- and RF-based classification it may have the best classification accuracy with most of the datasets.

6 Conclusion

This study focused to present the comparative result analysis and impact of feature section approaches with classification approaches to combine the benefits of feature selection with classifier. ILFS-, ECFS-, Relief-, FSV-, LS-, MFCS-, UDFS-, LLCFS- and CFS-based feature selection with various classification approaches such as LDA, NB, TDNN, KNN, DT, ELM, ELM-1, ELM-2, RF, PART, MLP, RBFN, SVM-R, SVM-P, LRA and SMO are utilized. For validation, seven real-world credit scoring datasets, namely Australian, Bank-marketing, Bankruptcy, Japanese, German categorical & numerical and Taiwan, are utilized and results are compared in terms of classification accuracy, sensitivity, specificity and G-measure. Comparative result analysis of various state-of-the-art approaches for credit score prediction models is also presented on this study and these approaches are categorized as classification, rule-based classification, ensemble and hybrid models.

From the literature, it is observed that in case of classification approaches SVM-L and SVM-R have best classification accuracy with Australian and German categorical datasets, respectively. With rule-based classification approaches, Multi-Objective GA-based fuzzy classification rule extraction has the best classification accuracy with Australian and Japanese datasets. With hybrid and ensemble approaches, NRS-based feature selection and layered ensemble classification approach with weighted voting as aggregation approach has the best classification accuracy with Australian and German categorical datasets. As in this study we have applied various classification with various feature selection, from obtained results, it is observed that TDNN has the best and RF has the second best approaches with most of the credit scoring datasets. And, CFS- or LLCFS-based feature selection approaches are more efficient and improve the performances of both classifiers most of the cases. Overall with all most classification approaches, UDFS has best performances with all most aforementioned classifiers and it also improves the classification performances as compared to performances with all features.

The results of this study with results of state-of-the-art approaches for credit scoring models are compared. As in this study, we have applied hybrid approach (Feature selection + classification), the comparative results from this study with TDNN have the best performances with all aforementioned credit scoring datasets as compared with model-based classification and hybrid approaches. Overall, it is the second best performer and NRS-based feature selection and layered ensemble classification is the best performer on credit scoring datasets. Further, applying the ensemble classification approaches with consideration of feature selection may have the best classification accuracy with most of the datasets.

References

1. Abdou, H.A., Pointon, J.: Credit scoring, statistical techniques and evaluation criteria: a review of the literature. Intell. Syst. Account. Finance Manag. **18**(2–3), 59–88 (2011)
2. Abellán, J., Castellano, J.G.: A comparative study on base classifiers in ensemble methods for credit scoring. Expert Syst. Appl. **73**, 1–10 (2017)
3. Ala'raj, M., Abbod, M.F.: Classifiers consensus system approach for credit scoring. Knowl. Based Syst. **104**, 89–105 (2016)
4. Ala'raj, M., Abbod, M.F.: A new hybrid ensemble credit scoring model based on classifiers consensus system approach. Expert Syst. Appl. **64**, 36–55 (2016)
5. Altman, N.S.: An introduction to kernel and nearest-neighbor nonparametric regression. Am. Stat. **46**(3), 175–185 (1992)
6. Atiya, A.F.: Bankruptcy prediction for credit risk using neural networks: a survey and new results. IEEE Trans. Neural Netw. **12**(4), 929–935 (2001)
7. Bashir, S., Qamar, U., Khan, F.H.: IntelliHealth: a medical decision support application using a novel weighted multi-layer classifier ensemble framework. J. Biomed. Inform. **59**, 185–200 (2016)
8. Bashir, S., Qamar, U., Khan, F.H., Naseem, L.: HMV: a medical decision support framework using multi-layer classifiers for disease prediction. J. Comput. Sci. **13**, 10–25 (2016)
9. Bequé, A., Lessmann, S.: Extreme learning machines for credit scoring: an empirical evaluation. Expert Syst. Appl. **86**, 42–53 (2017)
10. Bradley, P.S., Mangasarian, O.L.: Feature selection via concave minimization and support vector machines. ICML **98**, 82–90 (1998)
11. Breiman, L.: Random forests. Mach. Learn. **45**(1), 5–32 (2001)
12. Broomhead, D.S., Lowe, D.: Radial basis functions, multi-variable functional interpolation and adaptive networks. Technical report. Royal Signals and Radar Establishment Malvern (United Kingdom) (1988)
13. Cai, D., Zhang, C., He, X.: Unsupervised feature selection for multi-cluster data. In: Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 333–342. ACM (2010)
14. Chakravarthy, H., Bachan, P., Roshini, P., Ch, R.K.: Bio inspired approach as a problem solving technique (2012)
15. Chen, W., Ma, C., Ma, L.: Mining the customer credit using hybrid support vector machine technique. Expert Syst. Appl. **36**(4), 7611–7616 (2009)

16. Chi, B.W., Hsu, C.C.: A hybrid approach to integrate genetic algorithm into dual scoring model in enhancing the performance of credit scoring model. Expert Syst. Appl. **39**(3), 2650–2661 (2012)
17. Cortes, C., Vapnik, V.: Support-vector networks. Mach. Learn. **20**(3), 273–297 (1995)
18. Desai, V.S., Crook, J.N., Overstreet, G.A., Jr.: A comparison of neural networks and linear scoring models in the credit union environment. Eur. J. Oper. Res. **95**(1), 24–37 (1996)
19. Dua, D., Graff, C.: UCI machine learning repository. https://archive.ics.uci.edu/ml/machine-learning-databases/credit-screening/ (2017)
20. Duda, R.O., Hart, P.E., Stork, D.G.: Pattern Classification. Wiley, Hoboken (2012)
21. Edla, D.R., Tripathi, D., Cheruku, R., Kuppili, V.: An efficient multi-layer ensemble framework with BPSOGSA-based feature selection for credit scoring data analysis. Arab. J. Sci. Eng. **43**(12), 6909–6928 (2018)
22. Fisher, R.A.: The use of multiple measurements in taxonomic problems. Ann. Eugen. **7**(2), 179–188 (1936)
23. Frank, E., Witten, I.H.: Generating accurate rule sets without global optimization (1998)
24. Friedman, J., Hastie, T., Tibshirani, R.: The Elements of Statistical Learning. Springer Series in Statistics, vol. 1. Springer, New York (2001)
25. Fulwari, A.: Issues of housing finance in urban India a symptomatic study. Ph.D. thesis (2013)
26. Gorzałczany, M.B., Rudziński, F.: A multi-objective genetic optimization for fast, fuzzy rule-based credit classification with balanced accuracy and interpretability. Appl. Soft Comput. **40**, 206–220 (2016)
27. Green, S., Salkind, N.: Using SPSS for Windows and Macintosh: Analyzing and Understanding Data. Prentice Hall, Uppersaddle River (2010)
28. Guo, S., He, H., Huang, X.: A multi-stage self-adaptive classifier ensemble model with application in credit scoring. IEEE Access **7**, 78549–78559 (2019)
29. Hall, M.A.: Correlation-based feature selection for machine learning (1999)
30. Hastie, T., Tibshirani, R., Friedman, J.: The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Springer, Berlin (2009)
31. Hayashi, Y.: Application of a rule extraction algorithm family based on the Re-RX algorithm to financial credit risk assessment from a pareto optimal perspective. Oper. Res. Perspect. **3**, 32–42 (2016)
32. Haykin, S.: Neural Networks: A Comprehensive Foundation. Prentice Hall, Upper Saddle River (1994)
33. Haykin, S.S.: Neural Networks: A Comprehensive Foundation. Tsinghua University Press, Beijing (2001)
34. He, X., Cai, D., Niyogi, P.: Laplacian score for feature selection. In: Advances in Neural Information Processing Systems, pp. 507–514 (2006)
35. Hens, A.B., Tiwari, M.K.: Computational time reduction for credit scoring: an integrated approach based on support vector machine and stratified sampling method. Expert Syst. Appl. **39**(8), 6774–6781 (2012)
36. Hoffmann, F., Baesens, B., Mues, C., Van Gestel, T., Vanthienen, J.: Inferring descriptive and approximate fuzzy rules for credit scoring using evolutionary algorithms. Eur. J. Oper. Res. **177**(1), 540–555 (2007)
37. Hu, Q., Yu, D., Liu, J., Wu, C.: Neighborhood rough set based heterogeneous feature subset selection. Inf. Sci. **178**(18), 3577–3594 (2008)
38. Hu, Z., Bao, Y., Xiong, T., Chiong, R.: Hybrid filter-wrapper feature selection for short-term load forecasting. Eng. Appl. Artif. Intell. **40**, 17–27 (2015)
39. Huang, C.L., Chen, M.C., Wang, C.J.: Credit scoring with a data mining approach based on support vector machines. Expert Syst. Appl. **33**(4), 847–856 (2007)
40. Huang, C.L., Dun, J.F.: A distributed PSO-SVM hybrid system with feature selection and parameter optimization. Appl. Soft Comput. **8**(4), 1381–1391 (2008)
41. Huang, C.L., Wang, C.J.: A GA-based feature selection and parameters optimization for support vector machines. Expert Syst. Appl. **31**(2), 231–240 (2006)
42. Huang, G.B., Chen, L.: Convex incremental extreme learning machine. Neurocomputing **70**(16), 3056–3062 (2007)
43. Huang, G.B., Chen, L.: Enhanced random search based incremental extreme learning machine. Neurocomputing **71**(16), 3460–3468 (2008)
44. Huang, G.B., Wang, D.H., Lan, Y.: Extreme learning machines: a survey. Int. J. Mach. Learn. Cybernet. **2**(2), 107–122 (2011)
45. Huang, G.B., Zhou, H., Ding, X., Zhang, R.: Extreme learning machine for regression and multiclass classification. IEEE Trans. Syst. Man Cybern. Part B (Cybern.) **42**(2), 513–529 (2012)
46. Huang, G.B., Zhu, Q.Y., Siew, C.K.: Extreme learning machine: a new learning scheme of feedforward neural networks. In: Proceedings of 2004 IEEE International Joint Conference on of Neural Networks, vol. 2, pp. 985–990. IEEE (2004)
47. Jimbo Santana, P., Villa Monte, A., Rucci, E., Lanzarini, L.C., Fernández Bariviera, A.: Analysis of methods for generating classification rules applicable to credit risk. J. Comput. Sci. Technol. **17**, 20–28 (2017)
48. John, G.H., Langley, P.: Estimating continuous distributions in bayesian classifiers. In: Proceedings of the Eleventh Conference on Uncertainty in Artificial Intelligence, pp. 338–345. Morgan Kaufmann Publishers Inc. (1995)
49. Kala, R., Vazirani, H., Khanwalkar, N., Bhattacharya, M.: Evolutionary radial basis function network for classificatory problems. IJCSA **7**(4), 34–49 (2010)
50. Kang, S., Cho, S., Kang, P.: Multi-class classification via heterogeneous ensemble of one-class classifiers. Eng. Appl. Artif. Intell. **43**, 35–43 (2015)
51. Kaynak, C., Alpaydin, E.: Multistage cascading of multiple classifiers: one man's noise is another man's data. In: ICML, pp. 455–462 (2000)
52. Kim, M.J., Kang, D.K., Kim, H.B.: Geometric mean based boosting algorithm with over-sampling to resolve data imbalance problem for bankruptcy prediction. Expert Syst. Appl. **42**(3), 1074–1082 (2015)
53. Kira, K., Rendell, L.A.: The feature selection problem: traditional methods and a new algorithm. Aaai **2**, 129–134 (1992)
54. Kira, K., Rendell, L.A.: A practical approach to feature selection. In: Machine Learning Proceedings 1992, pp. 249–256. Elsevier (1992)
55. Kuppili, V., Tripathi, D., Reddy Edla, D.: Credit score classification using spiking extreme learning machine. Comput. Intell. **36**(2), 402–426 (2020)
56. Lahsasna, A., Ainon, R.N., Teh, Y.W.: Credit scoring models using soft computing methods: a survey. Int. Arab J. Inf. Technol. **7**(2), 115–123 (2010)
57. Lang, K.J.: A time-delay neural network architecture for speech recognition. Technical Report (1988)
58. Le Cessie, S., Van Houwelingen, J.C.: Ridge estimators in logistic regression. J. Appl. Stat. **41**, 191–201 (1992)
59. Lee, T.S., Chen, I.F.: A two-stage hybrid credit scoring model using artificial neural networks and multivariate adaptive regression splines. Expert Syst. Appl. **28**(4), 743–752 (2005)
60. Lee, T.S., Chiu, C.C., Lu, C.J., Chen, I.F.: Credit scoring using the hybrid neural discriminant technique. Expert Syst. Appl. **23**(3), 245–254 (2002)

61. Li, S.T., Shiue, W., Huang, M.H.: The evaluation of consumer loans using support vector machines. Expert Syst. Appl. **30**(4), 772–782 (2006)
62. Liang, D., Tsai, C.F., Dai, A.J., Eberle, W.: A novel classifier ensemble approach for financial distress prediction. Knowl. Inf. Syst. **54**, 437–462 (2017)
63. Liang, D., Tsai, C.F., Wu, H.T.: The effect of feature selection on financial distress prediction. Knowl.-Based Syst. **73**, 289–297 (2015)
64. Lichman, M.: UCI machine learning repository (2013). http://archive.ics.uci.edu/ml
65. Lin, W.Y., Hu, Y.H., Tsai, C.F.: Machine learning in financial crisis prediction: a survey. IEEE Trans. Syst. Man Cybern. Part C (Appl. Rev.) **42**(4), 421–436 (2012)
66. Liu, Y., Wang, G., Chen, H., Dong, H., Zhu, X., Wang, S.: An improved particle swarm optimization for feature selection. J. Bionic Eng. **8**(2), 191–200 (2011)
67. Louzada, F., Ara, A., Fernandes, G.B.: Classification methods applied to credit scoring: systematic review and overall comparison. Surv. Oper. Res. Manag. Sci. **21**, 117–134 (2016)
68. Martens, D., Baesens, B., Van Gestel, T., Vanthienen, J.: Comprehensible credit scoring models using rule extraction from support vector machines. Eur. J. Oper. Res. **183**(3), 1466–1476 (2007)
69. Martens, D., De Backer, M., Haesen, R., Vanthienen, J., Snoeck, M., Baesens, B.: Classification with ant colony optimization. IEEE Trans. Evol. Comput. **11**(5), 651–665 (2007)
70. Mester, L.J., et al.: What's the point of credit scoring? Bus. Rev. **3**(Sep/Oct), 3–16 (1997)
71. Moretti, F., Pizzuti, S., Panzieri, S., Annunziato, M.: Urban traffic flow forecasting through statistical and neural network bagging ensemble hybrid modeling. Neurocomputing **167**, 3–7 (2015)
72. Moro, S., Cortez, P., Rita, P.: A data-driven approach to predict the success of bank telemarketing. Decis. Support Syst. **62**, 22–31 (2014)
73. Nanni, L., Lumini, A.: An experimental comparison of ensemble of classifiers for bankruptcy prediction and credit scoring. Expert Syst. Appl. **36**(2), 3028–3033 (2009)
74. Neumann, F., Witt, C.: Bioinspired Computation in Combinatorial Optimization: Algorithms and Their Computational Complexity. Springer, Berlin (2010)
75. Oreski, S., Oreski, G.: Genetic algorithm-based heuristic for feature selection in credit risk assessment. Expert Syst. Appl. **41**(4), 2052–2064 (2014)
76. Paleologo, G., Elisseeff, A., Antonini, G.: Subagging for credit scoring models. Eur. J. Oper. Res. **201**(2), 490–499 (2010)
77. Parvin, H., MirnabiBaboli, M., Alinejad-Rokny, H.: Proposing a classifier ensemble framework based on classifier selection and decision tree. Eng. Appl. Artif. Intell. **37**, 34–42 (2015)
78. Ping, Y., Yongheng, L.: Neighborhood rough set and SVM based hybrid credit scoring classifier. Expert Syst. Appl. **38**(9), 11300–11304 (2011)
79. Platt, J.C.: 12 fast training of support vector machines using sequential minimal optimization. In: Advances in Kernel Methods, pp. 185–208 (1999)
80. Polat, K., Güneş, S., Arslan, A.: A cascade learning system for classification of diabetes disease: generalized discriminant analysis and least square support vector machine. Expert Syst. Appl. **34**(1), 482–487 (2008)
81. Qualitative bankruptcy data set. https://archive.ics.uci.edu/ml/machine-learning-databases/00281/. Accessed 1 Oct 2019
82. Rifkin, R.M.: Everything old is new again: a fresh look at historical approaches in machine learning. Ph.D. thesis, Massachusetts Institute of Technology (2002)
83. Rodriguez, J.J., Kuncheva, L.I., Alonso, C.J.: Rotation forest: a new classifier ensemble method. IEEE Trans. Pattern Anal. Mach. Intell. **28**(10), 1619–1630 (2006)
84. Roffo, G., Melzi, S.: Features selection via eigenvector centrality. In: Proceedings of New Frontiers in Mining Complex Patterns (NFMCP 2016) (Oct 2016) (2016)
85. Roffo, G., Melzi, S.: Ranking to learn. In: International Workshop on New Frontiers in Mining Complex Patterns, pp. 19–35. Springer (2016)
86. Roffo, G., Melzi, S., Castellani, U., Vinciarelli, A.: Infinite latent feature selection: a probabilistic latent graph-based ranking approach. In: Computer Vision and Pattern Recognition (2017)
87. Rokach, L., Maimon, O.Z.: Data Mining with Decision Trees: Theory and Applications, vol. 69
88. Rosenblatt, F.: Principles of neurodynamics. Perceptrons and the theory of brain mechanisms. Technical report, Cornell Aeronautical Lab Inc., Buffalo, NY (1961)
89. Rudziński, F.: A multi-objective genetic optimization of interpretability-oriented fuzzy rule-based classifiers. Appl. Soft Comput. **38**, 118–133 (2016)
90. Rumelhart, D.E., Hinton, G.E., Williams, R.J.: Learning internal representations by error propagation. Technical report, California Univ San Diego La Jolla Inst for Cognitive Science (1985)
91. Saha, M.: Credit cards issued. http://www.thehindu.com/business/Industry/Credit-cards-issued-touch-24.5-million/article14378386.ece (2017). Accessed 1 Oct 2019
92. Schölkopf, B., Tsuda, K., Vert, J.P., Istrail, D.S., Pevzner, P.A., Waterman, M.S., et al.: Kernel Methods in Computational Biology. MIT Press, Cambridge (2004)
93. Senliol, B., Gulgezen, G., Yu, L., Cataltepe, Z.: Fast correlation based filter (FCBF) with a different search strategy. In: 23rd International Symposium on Computer and Information Sciences, 2008. ISCIS'08, pp. 1–4. IEEE (2008)
94. Shahani, K., Udpa, L., Udpa, S.: Time delay neural networks for classification of ultrasonic NDT signals. In: Review of Progress in Quantitative Nondestructive Evaluation, pp. 693–700. Springer (1992)
95. Shukla, A.K., Singh, P., Vardhan, M.: A two-stage gene selection method for biomarker discovery from microarray data for cancer classification. Chemometr. Intell. Lab. Syst. **183**, 47–58 (2018)
96. Shukla, A.K., Tripathi, D.: Detecting biomarkers from microarray data using distributed correlation based gene selection. Genes Genomics **42**, 449–465 (2020)
97. Shukla, A.K., Tripathi, D., Reddy, B.R., Chandramohan, D.: A study on metaheuristics approaches for gene selection in microarray data: algorithms, applications and open challenges. Evolut. Intell. **13**, 309–329 (2019)
98. Statlog: Australian credit approval data set. http://archive.ics.uci.edu/ml/machine-learning-databases/statlog/australian/australian.dat. Accessed 1 Oct 2019
99. Statlog: German dataset. https://archive.ics.uci.edu/ml/machine-learning-databases/statlog/german/. Accessed 1 Oct 2019
100. Thomas, L.C., Edelman, D.B., Crook, J.N.: Credit Scoring and Its Applications. SIAM, Philadelphia (2002)
101. Tripathi, D., Cheruku, R., Bablani, A.: Relative performance evaluation of ensemble classification with feature reduction in credit scoring datasets. In: Reddy Edla, D., Lingras, P., Venkatanareshbabu, K. (eds.) Advances in Machine Learning and Data Science, pp. 293–304. Springer, Berlin (2018)
102. Tripathi, D., Edla, D.R., Cheruku, R.: Hybrid credit scoring model using neighborhood rough set and multi-layer ensemble classification. J. Intell. Fuzzy Syst. **34**(3), 1543–1549 (2018)
103. Tripathi, D., Edla, D.R., Cheruku, R., Kuppili, V.: A novel hybrid credit scoring model based on ensemble feature selection and multilayer ensemble classification. Comput. Intell. **35**, 371–394 (2019)
104. Tripathi, D., Edla, D.R., Kuppili, V., Bablani, A.: Evolutionary extreme learning machine with novel activation function for credit scoring. Eng. Appl. Artif. Intell. **96**, 103980 (2020)

105. Tripathi, D., Edla, D.R., Kuppili, V., Bablani, A., Dharavath, R.: Credit scoring model based on weighted voting and cluster based feature selection. Procedia Comput. Sci. **132**, 22–31 (2018)
106. Tripathi, D., Edla, D.R., Kuppili, V., Dharavath, R.: Binary BAT algorithm and RBFN based hybrid credit scoring model. Multimed. Tools Appl. **79**(43), 31889–31912 (2020)
107. Tsai, C.F.: Feature selection in bankruptcy prediction. Knowl.-Based Syst. **22**(2), 120–127 (2009)
108. Tsai, C.F., Wu, J.W.: Using neural network ensembles for bankruptcy prediction and credit scoring. Expert Syst. Appl. **34**(4), 2639–2649 (2008)
109. Van Gestel, T., Baesens, B., Suykens, J.A., Van den Poel, D., Baestaens, D.E., Willekens, M.: Bayesian kernel based classification for financial distress detection. Eur. J. Oper. Res. **172**(3), 979–1003 (2006)
110. Vapnik, V.: The Nature of Statistical Learning Theory. Springer, Berlin (2013)
111. Vapnik, V.N.: An overview of statistical learning theory. IEEE Trans. Neural Netw. **10**(5), 988–999 (1999)
112. Waibel, A., Hanazawa, T., Hinton, G., Shikano, K., Lang, K.J.: Phoneme recognition using time-delay neural networks. In: Readings in speech recognition, pp. 393–404. Elsevier (1990)
113. Wang, G., Hao, J., Ma, J., Jiang, H.: A comparative assessment of ensemble learning for credit scoring. Expert Syst. Appl. **38**(1), 223–230 (2011)
114. Wang, G., Ma, J., Huang, L., Xu, K.: Two credit scoring models based on dual strategy ensemble trees. Knowl.-Based Syst. **26**, 61–68 (2012)
115. Wang, J., Guo, K., Wang, S.: Rough set and Tabu search based feature selection for credit scoring. Procedia Comput. Sci. **1**(1), 2425–2432 (2010)
116. West, D.: Neural network credit scoring models. Comput. Oper. Res. **27**(11), 1131–1152 (2000)
117. Witten, I.H., Frank, E., Hall, M.A., Pal, C.J.: Data Mining: Practical Machine Learning Tools and Techniques. Morgan Kaufmann, Burlington (2016)
118. Wongchinsri, P., Kuratach, W.: Sr-based binary classification in credit scoring. In: 2017 14th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON), pp. 385–388. IEEE (2017)
119. Xia, Y., Liu, C., Da, B., Xie, F.: A novel heterogeneous ensemble credit scoring model based on bstacking approach. Expert Syst. Appl. **93**, 182–199 (2018)
120. Xiao, W.B., Fei, Q.: A study of personal credit scoring models on support vector machine with optimal choice of kernel function parameters. Syst. Eng. Theory Pract. **10**, 010 (2006)
121. Xu, D., Zhang, X., Feng, H.: Generalized fuzzy soft sets theory-based novel hybrid ensemble credit scoring model. Int. J. Finance Econ. **24**(2), 903–921 (2019)
122. Yang, Y.: Adaptive credit scoring with kernel learning methods. Eur. J. Oper. Res. **183**(3), 1521–1536 (2007)
123. Yang, Y., Shen, H.T., Ma, Z., Huang, Z., Zhou, X.: l2, 1-norm regularized discriminative feature selection for unsupervised learning. In: IJCAI Proceedings-international Joint Conference on Artificial Intelligence, vol. 22, p. 1589 (2011)
124. Yeh, I.C., Lien, C.H.: The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. Expert Syst. Appl. **36**(2), 2473–2480 (2009)
125. Yu, L., Wang, S., Lai, K.K.: An intelligent-agent-based fuzzy group decision making model for financial multicriteria decision support: the case of credit scoring. Eur. J. Oper. Res. **195**(3), 942–959 (2009)
126. Zeng, H., Cheung, Y.M.: Feature selection and kernel learning for local learning-based clustering. IEEE Trans. Pattern Anal. Mach. Intell. **33**(8), 1532–1547 (2011)
127. Zhang, D., Zhou, X., Leung, S.C., Zheng, J.: Vertical bagging decision trees model for credit scoring. Expert Syst. Appl. **37**(12), 7838–7843 (2010)
128. Zhang, W., He, H., Zhang, S.: A novel multi-stage hybrid model with enhanced multi-population niche genetic algorithm: an application in credit scoring. Expert Syst. Appl. **121**, 221–232 (2019)
129. Zhou, L., Lai, K.K., Yen, J.: Credit scoring models with AUC maximization based on weighted SVM. Int. J. Inf. Technol. Decis. Mak. **8**(04), 677–696 (2009)